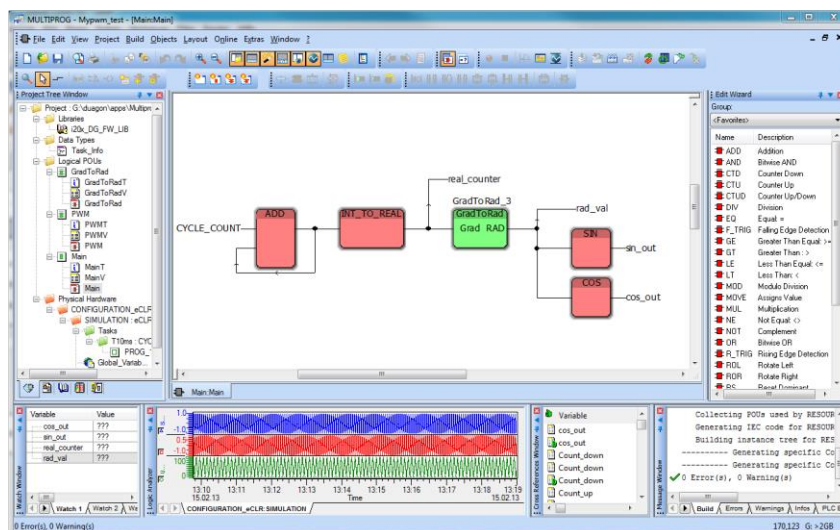




ionia™ IEC 61131 API

Data Sheet

This document describes the API used to create ionia™ application programs in an IEC 61131 style development environment using the MULTIPROG Express.



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3	2013-09-06	Dent	2013-09-23	Eberli		d-006935-014633
4	2014-07-09	Bollinger	2014-07-09	Eberli		d-006935-015200
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6	2014-11-26	Eberli	2014-12-18	Eberli		d-006935-018956
7	2015-02-16	Meier	2015-03-05	Kaufmann		d-006935-020293
8	2015-03-13	Paschalakis	2015-03-17	Eberli		d-006935-020675
9	2015-03-30	Meier	2015-03-30	Meier		d-006935-020951
10	2015-04-04	Meier	2015-04-02	Meier		d-006935-021080
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1. Introduction

This document describes the function blocks (FBs) and IO-drivers that are available to program the ionia™ PLC.

Release Information

This document refers to the following versions:

Deliverable	Reference
MULTIPROG Express for ionia™ (w/o redundancy operation library).	d-007271-020555, standard Library (order code: i401)
MULTIPROG Express for ionia™ (including redundancy operation library).	d-010864-019050, (order code: i401 + i402)
MULTIPROG Express for ionia™ (including redundancy operation library).	d-010864-064840

Table 1: MULTIPROG Express Releases for ionia™

2. MULTIPROG Interface

2.1. Firmware Interfaces

Functions (FUs) and function blocks (FB) provide the application programming interface (API) to the various I/O modules and to non-cyclical transmitted data, such as the MVB message data. The FU's and FB's are included into MULTIPROG Express by means of adding libraries. For each module resp. data communication type a specific library exists.

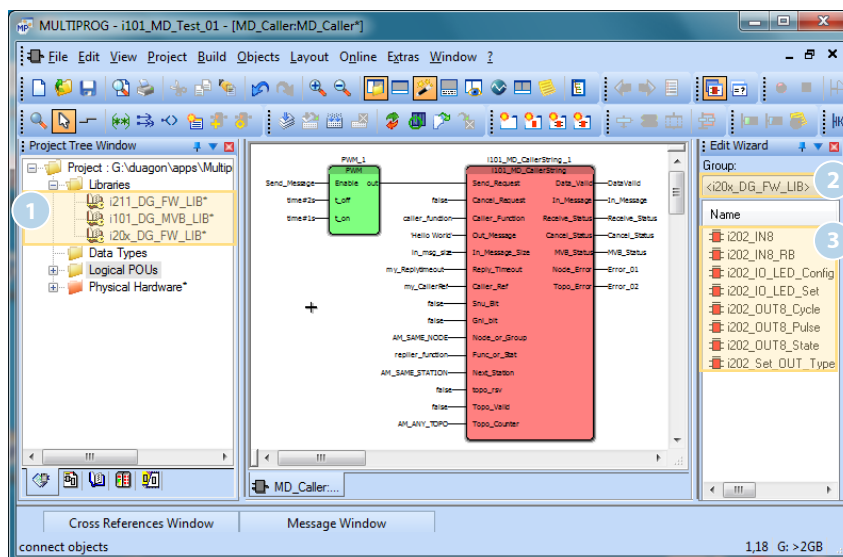


Figure 1: MULTIPROG Express main window overview

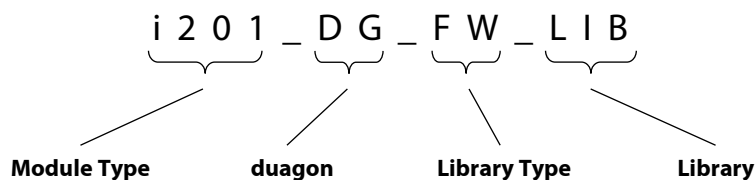
- 1 Libraries are included into MULTIPROG Express by means of a right click onto *Libraries* within the *Project Explorer*. The libraries can be stored anywhere on your local hard disk or on a network drive.

The libraries can be found in the installation directory of MULTIPROG Express (*MPE_INSTALLDIR*):

%MPE_INSTALLDIR%\PLC\FW_LIB\

- 2 Through the *Edit Wizard*'s filter option, it is possible to filter for a specific library.
- 3 According to the selected filter only the FUs and FBs belonging to the selected module are shown.

2.1.1. Naming Convention



Module Type:

- i101 MVB CPU module
- i102 Ethernet CPU module
- i201 Digital input module
- i202 Digital input / output module
- i205 Digital relay output module
- i211 Analog input module

- i213 Analog output module
- i301 MVB interface module
- i303 Ethernet interface module
- i305 Serial interface module
- i306 WTB interface module
- ionia Not module specific

Library Type:

- FW Functions and function blocks
- MVB MVB message data function blocks
- SER Serial communication function blocks
- UDP Ethernet and UDP communication function blocks
- WTB WTB communication function blocks

2.1.2. Library vs. Firmware Compatibility Check

Two error cases are detected by the versioning system:

1. The application program uses a firmware library which is not available on the PLC.
 - a. Error indication pop-up window:

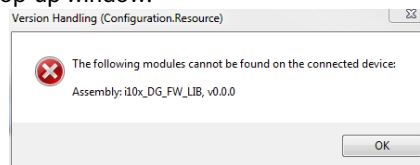


Figure 2: Error Pop-up – Missing Library

- b. Error indication within MULTIPROG Express:

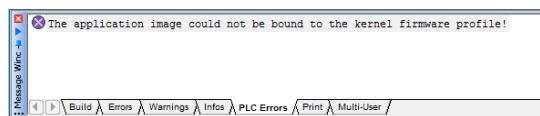


Figure 3: MULTIPROG Express Error– Missing Library

2. The application program uses a firmware library which is different to that running on the PLC.
 - a. Error indication pop-up window:

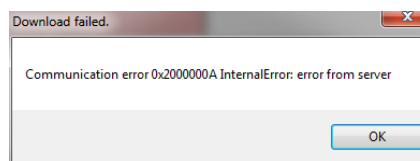


Figure 4: Error Pop-up – Incorrect Library Version

- b. Error indication within MULTIPROG Express:

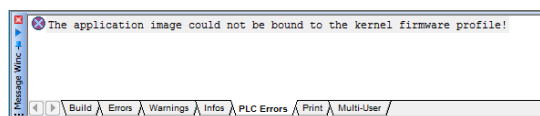


Figure 5: MULTIPROG Express Error– Incorrect Library Version

The error is raised at download time in both cases.

2.1.3. Library Versioning and Compatibility Handling

MOD_DG_FW_LIB_R1 designates Release 1 of the library *MOD_DG_FW_LIB*. If a new release of a library is loaded into an existing project, the FBs present in both versions

of the library are updated with the new version. Parameters which are present in both version of a function block remain connected in the logic.

2.2. I/O Driver Interface (Field Bus)

The interface to the field bus is provided by the so called I/O driver interface of MULTIPROG Express. The I/O configuration dialog is opened by means of a double click on the *IO_Configuration* icon. By pressing the push button *Add* at this dialog the appropriate field bus driver can be selected.

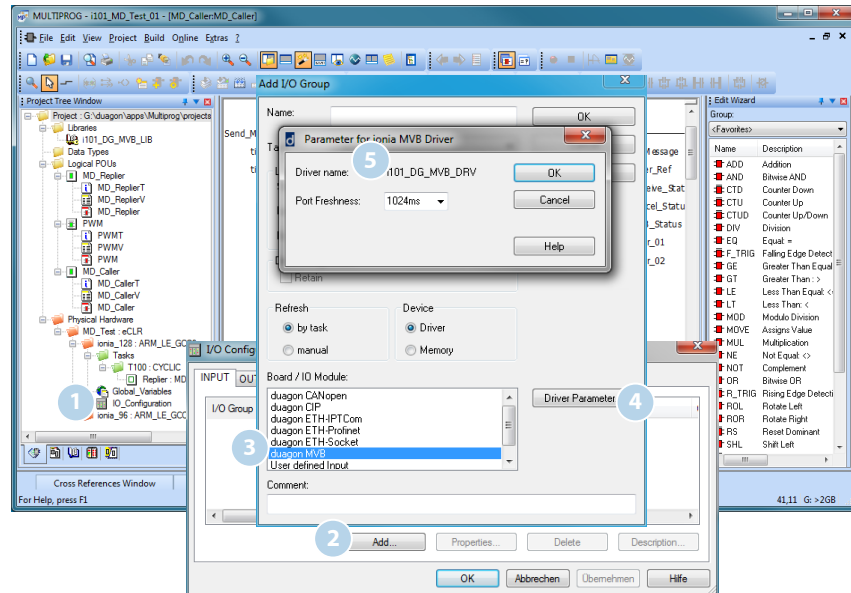
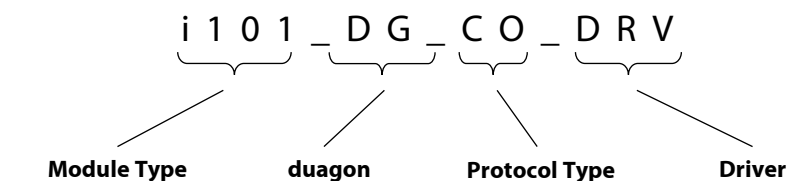


Figure 6: I/O Configuration Dialog (MULTIPROG Express)

- 1 Double click on the icon *IO_Configuration* opens the I/O Configuration dialog.
- 2 Press the *Add* button to open the *Add I/O Group* dialog.
- 3 Select the appropriate field bus.
- 4 By pressing the button *Driver Parameter*, the parameter dialog is opened.
- 5 Where the actual driver name is displayed and further settings can be carried out.

Note: For the i303 TRDP driver, select "user defined input", and under the category "Driver name", type "i303_DG_TRDP_DRV" for the MULTIPROG Express to identify driver from the eCLRDeviceProfiles XML configuration file and select the i303 TRDP as an I/O driver interface for the project.

Naming Convention



Module Type:

- i101 MVB CPU
- i103 CAN CPU

- i301 MVB interface module
- i303 Ethernet interface module
- ionia Not module specific

Protocol Type:

- MVB MVB driver for process data communication
- CO CANopen 2.0 driver for process data communication
- TRDP TRDP driver for process data communication

3. duagon Firmware Library

3.1. i201 Digital Input Module – i201_DG_FW_LIB Library

3.1.1. i201 Hardware Interface

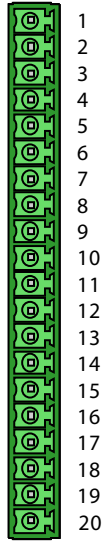


Figure 7: i201 I/O Connector

Pin #	Channel	Group (OP number < 18989)	Group (OP number >= 18989)	Meaning
1	-	-	-	Vbat+
2	-	-	-	Vbat-
3	1	1	-	IN
4	2	2	1	IN
5	3	1	-	IN
6	4	2	1	IN
7	5	1	-	IN
8	6	2	1	IN
9	7	1	-	IN
10	8	2	1	IN
11	-	-	-	IN
12	-	-	-	IN
13	9	1	-	IN
14	10	2	2	IN
15	11	1	-	IN
16	12	2	2	IN
17	13	1	-	IN
18	14	2	2	IN
19	15	1	-	IN
20	16	2	2	IN

Table 2: i201 Hardware Interface

3.1.2. i201 Software Interface

The following function blocks are included in the i201_DG_FW_LIB library:

Used Block	Purpose	OP Requirement	i201_IN16	i201_ENCIN8	i201_FIN4_G1	i201_FIN4_G2	i201_FIN8_G1	i201_FIN8_G2	i201_IO_LED_Config	i201_IO_LED_Set
i201_IN16	16 Channel digital input	-	-	-	-	-	-	-	-	-
i201_ENCIN8	8 Channel Period Counter and Direction Detection	>= d-nnnnn-018989	-	-	-	-	-	-	-	-
i201_FIN4_G1	4 Channel Frequency Input Group 1	>= d-nnnnn-018989	-	-	-	-	PR	PR	-	-
i201_FIN4_G2	4 Channel Frequency Input Group 2	>= d-nnnnn-018989	-	-	-	-	PR	PR	-	-
i201_FIN8_G1	8 Channel Frequency Input Group 1	< d-nnnnn-018989	-	-	PR	PR	-	-	-	-
i201_FIN8_G2	8 Channel Frequency Input Group 2	< d-nnnnn-018989	-	-	PR	PR	-	-	-	-
i201_IO_LED_Config	Set LED Conf	-	-	-	-	-	-	-	-	RE
i201_IO_LED_Set	Set LED State	-	-	-	-	-	-	-	MA	-

Figure 8: i201 Software Interface

When selecting function blocks for use on the same I/O module please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.1.3. i201_IN16

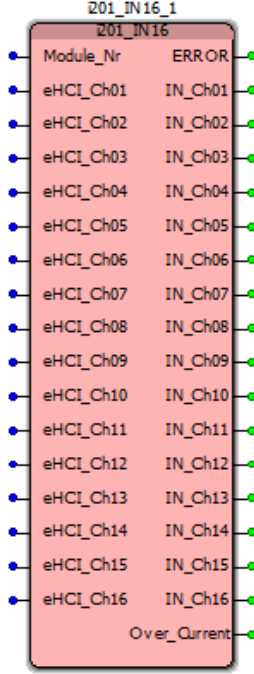
Name	i201_IN16	
Purpose	Get signals of the 16 digital inputs.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed..
	eHCI_Chxx	Enable high current input (HCI) mode for input channel xx.
Outputs	IN_Chxx	Indicates the state of input signal from channel xx.
	Over_Current	Reserved.
	ERROR	Error Code
Remarks	-	

Table 3: i201_IN16 Function Block Description

3.1.4. i201_ENCIN8

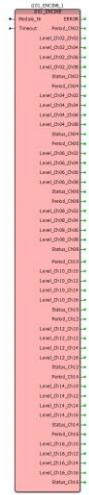
Name	i201_ENCIN8	
Purpose	Get period counter values and direction detection.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Timeout	Period timeout limit in microseconds (24-Bit Value).
Outputs	Period_Chxx	24-Bit period value in milliseconds of channel xx
	Level_Chxx_Chyy	1 Bit input level value of the other channels yy as channel xx to determine direction. For inputs I-2, 4, 6, 8 (upper 4 words of the data words): - Bit 0 I-2 - Bit 1 I-4 - Bit 2 I-6 - Bit 3 I-8 For inputs I-10, 12, 14, 16 (lower 4 words of the data words): - Bit 0 I-10 - Bit 1 I-12 - Bit 2 I-14 - Bit 3 I-16
	Status_Chxx	2 Bit status of the input channel xx word related to the period. 0 → OLD: Period not updated since last read 1 → NEW: Period is updated once, and valid since last read 2 → OVERWRITTEN: Period is updated multiple times since last read and therefore register is overwritten 3 → TIMEOUT: Timeout reached of period counter period not valid
	ERROR	Error Code
Remarks	Works only on FW number >= d-nnnnnn-018989.	

Table 4: i201_ENCIN8 Function Block Description

3.1.5. i201_FIN4_G1

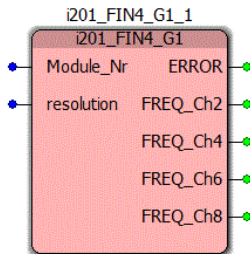
Name	i201_FIN4_G1	
Purpose	4 channel frequency counter input reading group 1.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 5: i201_FIN4_G1 Function Block Description

3.1.6. i201_FIN4_G2

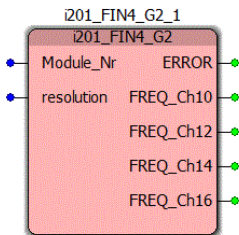
Name	i201_FIN4_G2	
Purpose	4 channel frequency counter input reading group 1.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 6: i201_FIN4_G2 Function Block Description

3.1.7. i201_FIN8_G1

Name	i201_FIN8_G1	
Purpose	8 channel frequency counter input reading group 1.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
Outputs	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 7: i201_FIN8_G1 Function Block Description

3.1.8. i201_FIN8_G2

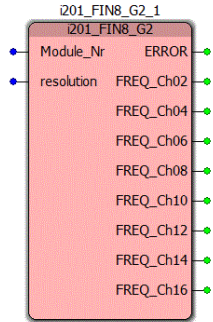
Name	i201_FIN8_G2	
Purpose	8 channel frequency counter input reading group 2.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
Outputs	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 8: i201_FIN8_G2 Function Block Description

3.1.9. i201_IO_LED_Config

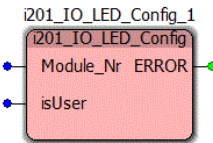
Name	i201_IO_LED_Config	
Purpose	Sets the LED's into user controlled mode or input controlled mode.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	isUser	TRUE: Sets the LED to be controlled by i201_LED_Set function FALSE: Sets the LES to be controlled by the state of its corresponding channel.
Outputs	ERROR	Error Code
Remarks	-	

Table 9: i201_IO_LED_Config function Block Description

3.1.10. i201_IO_LED_Set

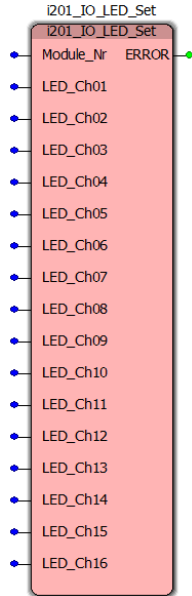
Name	i201_IO_LED_Set	
Purpose	Set the LED's ON or OFF on the 16 channels.	
Module type	i201	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	LED_Ch01 ... LED_Ch16	TRUE: Sets the LED ON FALSE: Sets the LED OFF.
Outputs	ERROR	Error Code
Remarks	-	

Table 10: i201_IO_LED_Set Function Block Description

3.2. i202 Digital In- / Output Module – i202_DG_FW_LIB Library

3.2.1. i202 Hardware Interface

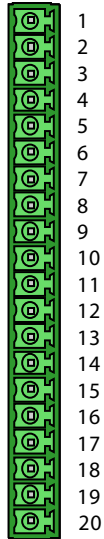


Figure 9: i202 I/O Connector

Pin #	Channel	Group (FW number < 18989)	Group (FW number >= 18989)	Meaning
1	-	-	-	Vbat+
2	-	-	-	Vbat-
3	1	1	-	IN / OUT
4	2	2	1	IN
5	3	1	-	IN / OUT
6	4	2	1	IN
7	5	1	-	IN / OUT
8	6	2	1	IN
9	7	1	-	IN / OUT
10	8	2	1	IN
11	-	-	-	IN
12	-	-	-	IN
13	9	1	-	IN / OUT
14	10	2	2	IN
15	11	1	-	IN / OUT
16	12	2	2	IN
17	13	1	-	IN / OUT
18	14	2	2	IN
19	15	1	-	IN / OUT
20	16	2	2	IN

Table 11: i202 Hardware Interface

3.2.2. i202 Software Interface

The following function blocks are included in the i202_DG_FW_LIB library:

Used Block	Purpose	OP Requirement	i202_IN0	i202_IN0_RB	i202_ENC0NB	i202_FTH0_C1	i202_FTH0_C2	i202_FTH0_C1	i202_FTH0_C2	i202_IO_LED_Config	i202_IO_LED_Set	i202_Out0_Cycle_C1	i202_Out0_Cycle_C2	i202_Out0_Pulse_C1	i202_Out0_Pulse_C2	i202_Out0_State
i202_IN0	8 Channel digital input	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
i202_IN0_RB	8 Channel digital input with input (I/O)	-	-	-	PR	PR	PR	PR	PR	-	-	PR	PR	PR	PR	RE
i202_ENC0NB	8 Channel Period Counter and Direction Detection	≠ 0-nominal-018989	-	-	-	-	-	-	-	-	-	-	-	-	-	-
i202_FTH0_G1	4 Channel Frequency Input Group 1	≠ 0-nominal-018989	-	PR	-	-	-	PR	PR	-	-	-	-	-	-	-
i202_FTH0_G2	4 Channel Frequency Input Group 2	≠ 0-nominal-018989	-	PR	-	-	-	PR	PR	-	-	-	-	-	-	-
i202_FTH0_G1	8 Channel Frequency Input Group 1	< 0-nominal-018989	-	PR	-	PR	PR	-	-	-	-	-	-	-	-	-
i202_FTH0_G2	8 Channel Frequency Input Group 2	< 0-nominal-018989	-	PR	-	PR	PR	-	-	-	-	-	-	-	-	-
i202_IO_LED_Config	Set LED Conf	-	-	-	-	-	-	-	-	RE	-	-	-	-	-	-
i202_IO_LED_Set	Set LED State	-	-	-	-	-	-	-	-	MA	-	-	-	-	-	-
i202_Out0_Cycle_G1	Set PWM Duty Cycle of Group 1	-	-	PR	-	-	-	-	-	-	-	-	-	PR	-	PR
i202_Out0_Cycle_G2	Set PWM Duty Cycle of Group 2	-	-	PR	-	-	-	-	-	-	-	-	-	PR	-	PR
i202_Out0_Pulse_G1	Set PWM Pulse Width of Group 1	-	-	PR	-	-	-	-	-	-	-	PR	-	-	-	PR
i202_Out0_Pulse_G2	Set PWM Pulse Width of Group 2	-	-	PR	-	-	-	-	-	-	-	PR	-	-	-	PR
i202_Out0_State	Set 8 digital outputs	-	-	RE	-	-	-	-	-	-	-	PR	PR	PR	PR	PR

Figure 10: i202 Software Interface

When selecting function blocks for use on the same I/O module please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.2.3. i202_IN8

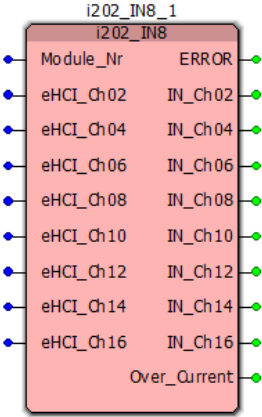
Name	i202_IN8	
Purpose	Get signals of the 8 digital inputs.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed..
	eHCI_Chxx	Enable high current input (HCI) mode for input channel xx of groups 1 and 2 (group 2 for FW number < 18989).
Outputs	IN_Chxx	Indicates the state of input signal from channel xx.
	Over_Current	Reserved.
	ERROR	Error Code
Remarks	-	

Table 12: i202_IN8 Function Block Description

3.2.4. i202_IN8

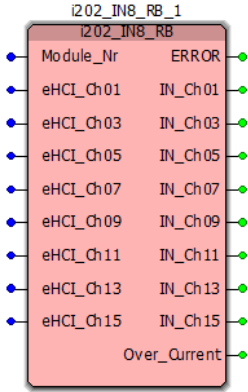
Name	i202_IN8_RB	
Purpose	Get signals of the 8 digital inputs (read-back inputs).	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed..
	eHCI_Chxx	Enable high current input (HCI) mode for input channel xx (group 1 for FW number < 18989).
Outputs	IN_Chxx	Indicates the state of input signal from channel xx.
	Over_Current	Over current of RB channels: Bit 0: IO-1 Bit 1: IO-3 ... Bit7: IO-15 Remark: Available for FW number >= d-nnnnnn-018989
	ERROR	Error Code
Remarks	-	

Table 13: i202_IN8_RB Function Block Description

3.2.5. i202_ENCIN8

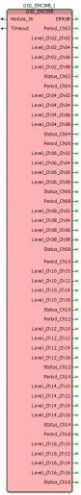
Name	i202_ENCIN8	
Purpose	Get period counter values and direction detection.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Timeout	Period timeout limit in microseconds (24-Bit Value).
Outputs	Period_Chxx	24-Bit period value in milliseconds of channel xx
	Level_Chxx_Chyy	1 Bit input level value of the other channels yy as channel xx to determine direction. For inputs I-2, 4, 6, 8 (upper 4 words of the data words): - Bit 0 I-2 - Bit 1 I-4 - Bit 2 I-6 - Bit 3 I-8 For inputs I-10, 12, 14, 16 (lower 4 words of the data words): - Bit 0 I-10 - Bit 1 I-12 - Bit 2 I-14 - Bit 3 I-16
	Status_Chxx	2 Bit status of the input channel xx word related to the period. 0 → OLD: Period not updated since last read 1 → NEW: Period is updated once, and valid since last read 2 → OVERWRITTEN: Period is updated multiple times since last read and therefore register is overwritten 3 → TIMEOUT: Timeout reached of period counter period not valid
	ERROR	Error Code
Remarks	Works only on FW number >= d-nnnnnn-018989.	

Table 14: i202_ENCIN8 Function Block Description

3.2.6. i202_FIN4_G1

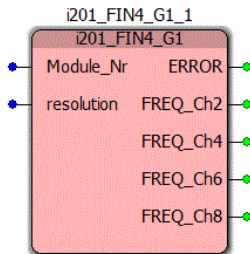
Name	i202_FIN4_G1	
Purpose	4 channel frequency counter input reading group 1.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 15: i202_FIN4_G1 Function Block Description

3.2.7. i202_FIN4_G2

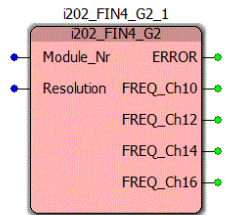
Name	i202_FIN4_G2	
Purpose	4 channel frequency counter input reading group 1.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 16: i202_FIN4_G2 Function Block Description

3.2.8. i202_FIN8_G1

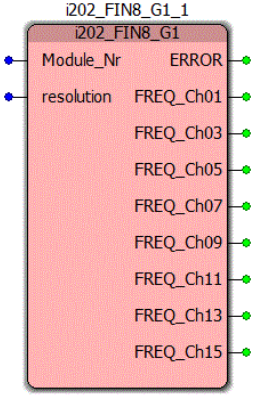
Name	i202_FIN8_G1	
Purpose	8 channel frequency counter input reading group 1.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
Outputs	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 17: i202_FIN8_G1 Function Block Description

3.2.9. i202_FIN8_G2

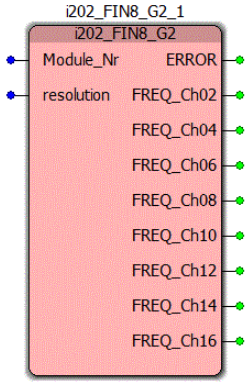
Name	i202_FIN8_G2	
Purpose	8 channel frequency counter input reading group 2.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	resolution	Set resolution time for group (1, 10, 100, 1000 [Hz]).
Outputs	FREQ_Chxx	Frequency on channel xx [Hz]
	ERROR	Error Code
Remarks	Works only on OP Number >= d-nnnnnn-018989.	

Table 18: i202_FIN8_G2 Function Block Description

3.2.10. i202_OUT8_Pulse_G1

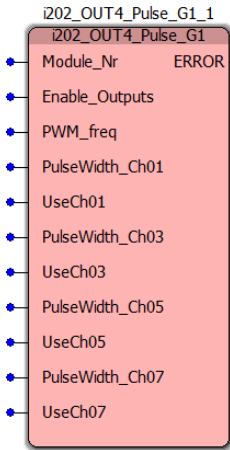
Name	i202_OUT8_Pulse_G1	
Purpose	Set the pulse width (in microseconds) for the corresponding output channel, when it is configured as a PWM output. [resolution 1 microsecond] (time based)	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Enable_Outputs	TRUE: (Used) output states reflect PulseWidth_Ch01 ... PulseWidth_Ch07 FALSE: all outputs disabled (low i.e. high impedance)
	PWM_freq	Base PWM Frequency of Group 1. 0 = PWM disabled (default) Frequency range: 25 ... 10000 Hz
	PulseWidth_Ch01 ... PulseWidth_Ch07	Pulse width in microseconds. [10us ... (T _p -10us)] 0 = PWM Channel disabled (default)
	UseCh01 ... UseCh07	TRUE: The channels is used to output the PWM frequency FALSE: The output is not used to output the PWM frequency. (The channel can be used with an i202_OUT8_State FB)
Outputs	ERROR	Error Code
Remarks	-	

Table 19: i202_OUT8_Pulse_G1 Function Block Description

3.2.11. i202_OUT8_Pulse_G1

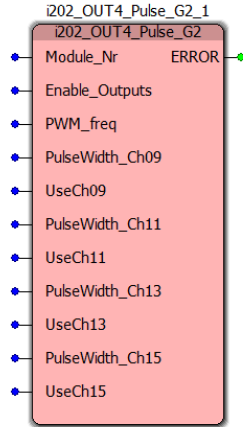
Name	i202_OUT8_Pulse_G1	
Purpose	Set the pulse width (in microseconds) for the corresponding output channel, when it is configured as a PWM output. [resolution 1 microsecond] (time based)	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Enable_Outputs	TRUE: (Used) output states reflect PulseWidth_Ch09 ... PulseWidth_Ch15 FALSE: all outputs disabled (low i.e. high impedance)
	PWM_freq	Base PWM Frequency of Group 2. 0 = PWM disabled (default) Frequency range: 25 ... 10000 Hz
	PulseWidth_Ch09 ... PulseWidth_Ch15	Pulse width in microseconds. [10us ... (Tp-10us)] 0 = PWM Channel disabled (default)
	UseCh09 ... UseCh15	TRUE: The channels is used to output the PWM frequency FALSE: The output is not used to output the PWM frequency. (The channel can be used with an i202_OUT8_State FB)
Outputs	ERROR	Error Code
Remarks	-	

Table 20: i202_OUT8_Pulse_G2 Function Block Description

3.2.12. i202_OUT4_Cycle_G1

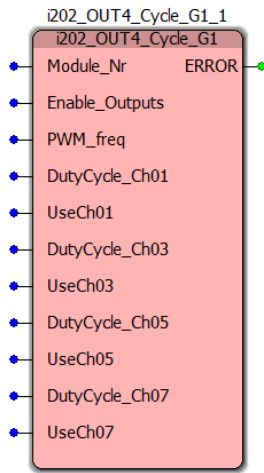
Name	i202_OUT4_Cycle_G1	
Purpose	Set the duty cycle in percent of the pulse width for an output channel, when it is configured as a PWM output. [resolution 1 %] (duty cycle based)	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Enable_Outputs	TRUE: (Used) output states reflect DutyCycle_Ch01 ... DutyCycle_Ch07. FALSE: all outputs disabled (low, i.e. high impedance).
	PWM_freq	Base PWM Frequency of Group 2. 0 = PWM disabled (default) Frequency range: 25 ... 10000 Hz
	DutyCycle_Ch01 ... DutyCycle_Ch07	Duty Cycle in % [range 0 ... 100] 0 = PWM Channel disabled (default).
	UseCh01 ... UseCh07	TRUE: The channels is used to output the PWM frequency FALSE: The output is not used to output the PWM frequency. (The channel can be used with an i202_OUT8_State FB)
Outputs	ERROR	Error Code
Remarks	-	

Table 21: i202_OUT4_Cycle_G1 Function Block Description

3.2.13. i202_OUT4_Cycle_G2

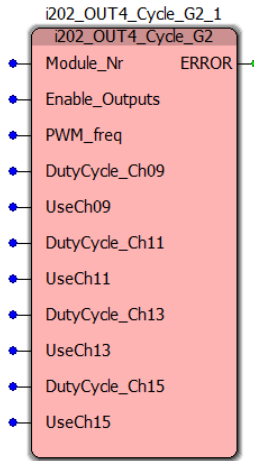
Name	i202_OUT4_Cycle_G2	
Purpose	Set the duty cycle in percent of the pulse width for an output channel, when it is configured as a PWM output. [resolution 1 %] (duty cycle based)	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Enable_Outputs	TRUE: (Used) output states reflect DutyCycle_Ch09 ... DutyCycle_Ch15. FALSE: all outputs disabled (low, i.e. high impedance).
	PWM_freq	Base PWM Frequency of Group 2. 0 = PWM disabled (default) Frequency range: 25 ... 10000 Hz
	DutyCycle_Ch09 ... DutyCycle_Ch15	Duty Cycle in % [range 0 ... 100] 0 = PWM Channel disabled (default).
	UseCh09 ... UseCh15	TRUE: The channels is used to output the PWM frequency FALSE: The output is not used to output the PWM frequency. (The channel can be used with an i202_OUT8_State FB)
Outputs	ERROR	Error Code
Remarks	-	

Table 22: i202_OUT4_Cycle_G1 Function Block Description

3.2.14. i202_OUT8_State

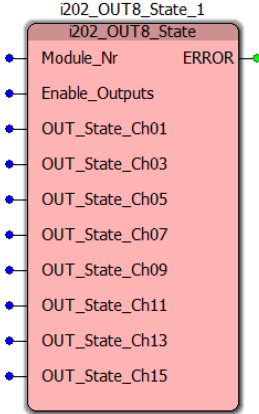
Name	i202_OUT8_State	
Purpose	Set the state of an output channel, when it is configured as a state output.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Enable_Outputs	TRUE: output states reflect OUT_Ch01 ... OUT_Ch15 FALSE: all outputs disabled (Low i.e. high impedance).
	OUT_State_Ch01 ... OUT_State_Ch15	TRUE: Channel output set HIGH. FALSE: Channel output set LOW.
Outputs	ERROR	Error Code
Remarks	Requires i202_Set_OUT_Type with Set_Type_Pulse_Chxx set to FALSE for each channel xx used as a state output. The Module_Nr numbers must match.	

Table 23: i202_OUT8_State Function Block Description

3.2.15. i202_IO_LED_Config

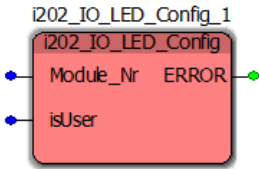
Name	i202_IO_LED_Config	
Purpose	Sets the LED's into user controlled mode or input controlled mode.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	isUser	TRUE: Sets the LED to be controlled by i201_LED_Set function FALSE: Sets the LES to be controlled by the state of its corresponding channel.
Outputs	ERROR	Error Code
Remarks	-	

Table 24: i202_IO_LED_Config Function Block Description

3.2.16. i202_IO_LED_Set

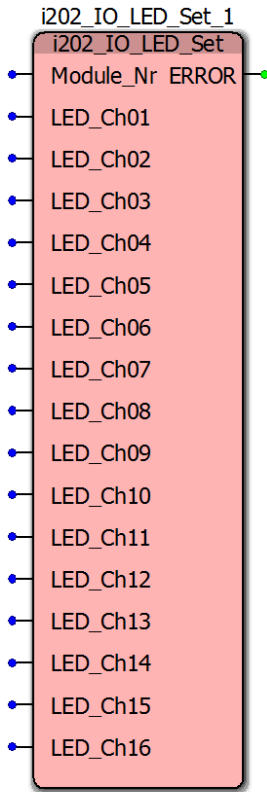
Name	i202_IO_LED_Set	
Purpose	Set the LED's ON or OFF on the 16 channels.	
Module type	i202	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	LED_Ch01 ... LED_Ch16	TRUE: Sets the LED ON FALSE: Sets the LED OFF.
Outputs	ERROR	Error Code
Remarks	-	

Table 25: i202_IO_LED_Set Function Block Description

3.3. i211 Analog Input – i211_DG_FW_LIB library

3.3.1. i211 Hardware Interface

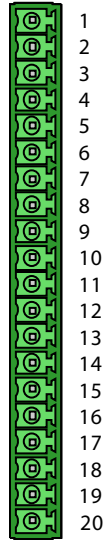


Figure 11: i211 I/O Connector

Pin #	Channel	Group
1	-	-
2	1	1
3	2	1
4	3	1
5	4	1
6	-	-
7	5	2
8	6	2
9	7	2
10	8	2
11	-	-
12	9	3
13	10	3
14	11	3
15	12	3
16	-	-
17	13	4
18	14	4
19	15	4
20	16	4

Table 26: i211 Hardware Interface

3.3.2. i211 Software Interface

The following function blocks are included in the i211_DG_FW_LIB library:

Used Block	Purpose	OP Requirement	i211_AIN_C	i211_AIN_R	i211_AIN_V	i211_AIN_PT100	i211_AIN_PT1000	i211_AIN_4W	i211_AIN_24	i211_AIN_LED_Config	i211_AIN_LED_Set
i211_AIN_C	3 channel current measurement input	-		PR	PR	PR	PR	PR	-	-	-
i211_AIN_R	3 channel resistance measurement input	-	PR		PR	PR	PR	PR	-	-	-
i211_AIN_V	3 channel voltage measurement input	-	PR	PR		PR	PR	PR	-	-	-
i211_AIN_PT100	4 Wire PT100 temperature sensor measurement	-	PR	PR	PR		-	-	-	-	-
i211_AIN_PT1000	4 Wire PT1000 temperature sensor measurement	-	PR	PR	PR	-		-	-	-	-
i211_AIN_4W	4 Wire resistance measurement	-	PR	PR	PR	-	-		-	-	-
i211_AIN_24	Get all raw ADC values	-	-	-	-	-	-	-		-	-
i211_AIN_LED_Config	Set LED Conf	-	-	-	-	-	-	-	-		RE
i211_AIN_LED_Set	Set LED State	-	-	-	-	-	-	-	-	MA	

Table 27: i211 Function Blocks Overview

When selecting function blocks for use on the same I/O module please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.3.3. i211_AIN_C

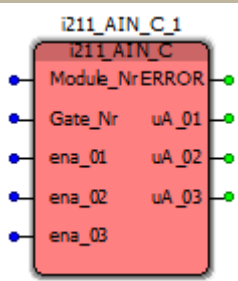
Name	i211_AIN_C	
Purpose	Input current measurement on specified group.	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Group_Nr	Group selection (1...4).
	ena_01 ... ena_03	0: Measurement on channel uA_01 ... uA_03 deactivated. 1: Measurement on channel uA_01 ... uA_03 activated.
Outputs	uA_01 ... uA_03	Process value in uA.
	ERROR	Error Code
Remarks	-	

Table 28: i211_AIN_C Function Block Description

3.3.4. i211_AIN_R

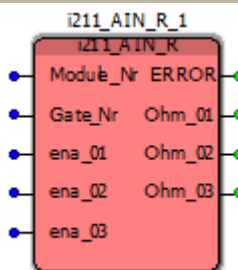
Name	i211_AIN_R	
Purpose	Resistive measurement on specified group.	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Group_Nr	Group selection (1...4).
	ena_01 ... ena_03	0: Measurement on channel Ohm_01 ... Ohm_03 deactivated. 1: Measurement on channel Ohm_01 ... Ohm_03 activated.
Outputs	Ohm_01 ... Ohm_03	Process value in Ohm.
	ERROR	Error Code
Remarks	-	

Table 29: i211_AIN_R Function Block Description

3.3.5. i211_AIN_V

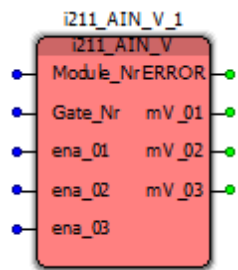
Name	i211_AIN_V	
Purpose	Voltage measurement on specified group.	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Group_Nr	Group selection (1...4).
	ena_01 ... ena_03	0: Measurement on channel mV_01 ... mV_03 deactivated. 1: Measurement on channel mV_01 ... mV_03 activated.
Outputs	mv_01 ... mv_03	Process value in mV.
	ERROR	Error Code
Remarks	-	

Table 30: i211_AIN_V Function Block Description

3.3.6. i211_AIN_PT100

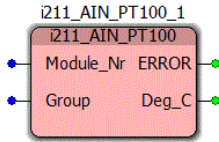
Name	i211_AIN_PT100	
Purpose	Four wire temperature measurement (with a PT100 temperature sensor).	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Group_Nr	Group selection (1...4).
	Deg_C	Process value in degrees Celsius.
	ERROR	Error Code
Remarks	-	

Table 31: i211_AIN_PT100 Function Block Description

3.3.7. i211_AIN_PT1000

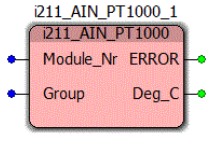
Name	i211_AIN_PT1000	
Purpose	Four wire temperature measurement (with a PT1000 temperature sensor).	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Group_Nr	Group selection (1...4).
	Deg_C	Process value in degrees Celsius.
	ERROR	Error Code
Remarks	-	

Table 32: i211_AIN_PT1000 Function Block Description

3.3.8. i211_AIN_4W

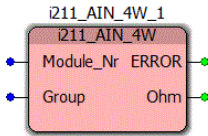
Name	i211_AIN_4W	
Purpose	Four wire temperature measurement (with a 4W temperature sensor).	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Group_Nr	Group selection (1...4).
Outputs	Deg_C	Process value in Ohms.
	ERROR	Error Code
Remarks	-	

Table 33: i211_AIN_4W Function Block Description

3.3.9. i211_AIN_24

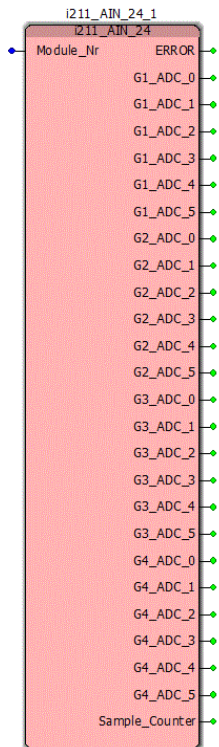
Name	i211_AIN_24	
Purpose	All ADC measurement values.	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
Outputs	G1_ADC_0 ... G4_ADC_5	Value delivered by the i211 for each ADC in raw form.
	Sample_Counter	Value increases at each sampling of the ADC's.
	ERROR	Error Code
Remarks	-	

Table 34: i211_AIN_24 Function Block Description

3.3.10. i211_LED_Config

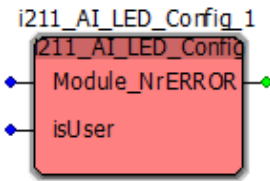
Name	i211_LED_Config	
Purpose	Sets the LED's into user controlled mode or input controlled mode.	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	isUser	TRUE: Sets the LED to be controlled by i211_AI_LED_Set function FALSE: Sets the LED to be controlled by the state of its corresponding channel.
Outputs	ERROR	Error Code
Remarks	-	

Table 35: i211_LED_Config Function Block Description

3.3.11. i211_AI_LED_Set

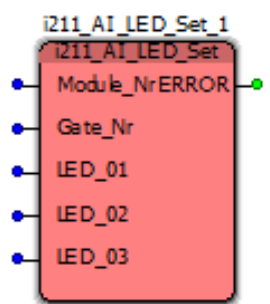
Name	i211_AI_LED_Set	
Purpose	Set the LED's ON or OFF on the 4 channels of a specific group.	
Module type	i211	
Symbol		
IEC 61131 type	Function block	
Parameters	Module_Nr	Indicates which I/O module in the ionia™ system is addressed.
	Group_Nr	Group selection (1...4).
	LED_Ch01 ... LED_Ch03	TRUE: Sets the LED ON FALSE: Sets the LED OFF.
Outputs	ERROR	Error Code
Remarks	-	

Table 36: i211_AI_LED_Set Function Block Description

3.4. i101/301 MVB Message Data Communication – i101/i301_DG_MVB_LIB Library

3.4.1. i101 Hardware Interface



Figure 12: i101 MVB and Service Connectors

Interface	Description
X1	MVB Interface 1
X2	MVB Interface 2
USB	USB Service Interface
Ethernet	Ethernet Service Interface

Table 37: i101 Hardware Interface

3.4.2. i301 Hardware Interface



Figure 13: i301 MVB and Service Connectors

Interface	Description
X1	MVB Interface 1
X2	MVB Interface 2

Table 38: i301 Hardware Interface

3.4.3. i101 Software Interface

The following function blocks are included in the i101_DG_MVB_LIB library:

Function	Purpose	Type	PR	MA	RE
i101_MD_CallerString	Send message	MSG data		X ¹	
i101_MD_ReplierString	Reply message	MSG data		X ¹	

Table 39: i101/i301 Function Block Overview

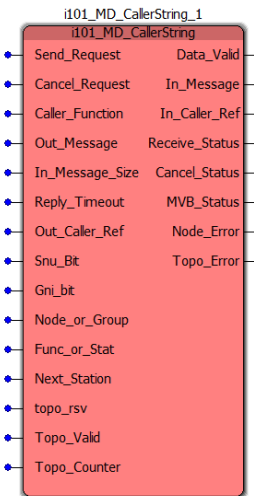
Remarks:

- 1) May used on different devices.

When selecting function blocks for use on the same communication channel please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.4.4. i101_MD_CallerString

Name	i101_MD_CallerString	
Purpose	Send a string to a replier and receive a confirmation string of the replier.	
Module type	i101	
Symbol		
IEC 61131 type	Function block	
Parameters	Send_Request	TURE: Call request is emitted. FALSE: Function block is disabled.
	Cancel_Request	Do not use.
	Caller_Function	Function ID of the caller.
	Out_Message	String to be sent to the message replier.
	In_Message_Size	Size of the expected reply message in Bytes.
	Reply_Timeout	Timeout value in multiples of 64 ms for the reply after the transfer of the Call_Message.

Outputs	Out_Caller_Ref	External reference for the call to be returned by the replier. It can be used in any way by the caller.
	Snu_Bit	System, not user Bit. TRUE: Message uses system addressing. FALSE: Indicates a User_Address
	Gni_Bit	Group, not Individual. TRUE: Indicates a group Address FALSE: Indicates an individual (node) address.
	Node_or_Group	Gni_Bit = FALSE: Specify a node Address (6 Bit) Gni_Bit = TRUE: Specify a group Address (6 Bit)
	Func_or_Stat	Snu_Bit = FALSE: Specify a function ID Snu_bit = TRUE: Specify a station ID
	Next_Station	Next Station ID
	topo_rsv	Reserved, always FALSE.
	Topo_Valid	Indicates if the following Topo_Counter is valid.
	Topo_Counter	Topo_Counter or AM_ANY_TOPO (=0).
	Data_Valid	TRUE: Received message is valid. FALSE: Received message is invalid resp. old.
	In_Message	Reply message (String) from the replier.
	In_Caller_Ref	Returned external reference from the replier.
	Receive_Status	Type: AM_RESULT Default value: AM_FAILURE
	Cancel_Status	Do not use.
	MVB_Status	Type: I101_ERROR_CODE Default value: AM_FAILURE
	Node_Error	TRUE: Input Node_or_Group outside range of 6-Bit FALSE: Input Node_or_Group within range of 6-Bit
	Topo_Error	TRUE: Input Topo_Counter outside range of 6-Bit FALSE: Input Topo_Counter within range of 6-Bit
Remarks	-	

Table 40: i101_MD_CallerString Function Block Description

3.4.5. i101_MD_ReplierString

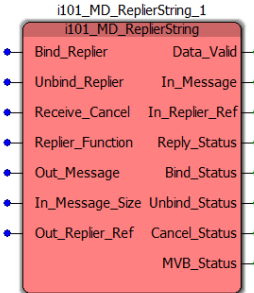
Name	i101_MD_ReplierString	
Purpose	Receive a string from the caller and confirm the message with a reply string.	
Module type	i101	
Symbol		
IEC 61131 type	Function block	
Parameters	Bind_Replier	TRUE: Makes a Replier known to the Messenger and connects its procedures. FALSE: Function block in disabled.
	Unbind_Replier	Do not use.
	Receive_Cancel	Do not use.
	Replier_Function	Function ID of the Replier expecting a call. Function ID = 253 implies a system address.
	Out_Message	String to be sent to the caller.
	In_Message_Size	Size of the expected caller string in Bytes.
	Out_Replier_Ref	External reference returned with the related receive_confirm procedure. It is at the same time an instance reference and distinguishes the replier instances which serve the same replier.
Outputs	Data_Valid	TRUE: Received message is valid. FALSE: Received message is invalid resp. old.
	In_Message	Called message (String) from the caller.
	In_Replier_Ref	Returned external reference from the caller.
	Reply_Status	Type: AM_RESULT Default value: AM_FAILURE
	Bind_Status	Type: AM_RESULT Default value: AM_FAILURE
	Unbind_Status	Do not use.
	Cancel_Status	Do not use.
	MVB_Status	Type: I101_ERROR_CODE Default value: AM_FAILURE
Remarks	-	

Table 41: i101_MD_ReplierString Function Block Description

3.5. i102 UDP Data Communication – i102_DG_UDP_LIB Library

3.5.1. i102 Hardware Interface

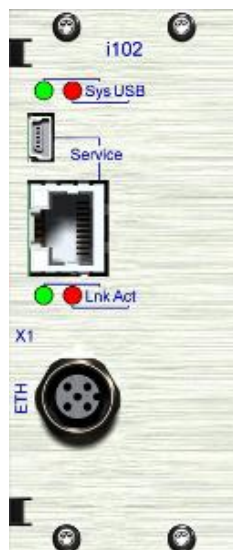


Figure 14: i102 Ethernet and Service Connectors

Interface	Description
X1	Ethernet Interface 1
USB	USB Service Interface
Ethernet	Ethernet Service Interface

Table 42: i102 Hardware Interface

3.5.2. i102 Software Interface

The following function blocks are included in the i102_DG_UDP_LIB library:

Function	Purpose	Type	PR	MA	RE
i102_UDP_ReceiverArray	Receive data	UDP data		X ¹	
i102_UDP_SenderArray	Send data	UDP data		X ¹	
ETH_Config	Set Ethernet configuration	control		X	
ETH_Get_Config	Get Ethernet configuration	control			

Table 43: i102 Function Blocks Overview

Remarks:

- 1) May used on different devices.

When selecting function blocks for use on the same communication channel please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.5.3. i102_UDP_ReceiverArray

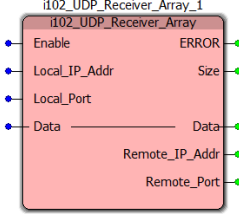
Name	i102_UDP_ReceiverArray	
Purpose	Used to receive an UDP datagram from a sender.	
Module type	i102	
Symbol		
IEC 61131 type	Function block	
Parameters	Enable	TRUE: Opens a port on the Ethernet interface as defined by its IPv4 address and begins to listen for UDP datagrams. FALSE: FB in disabled.
	Local_IP_addr	Denotes the IPv4 address of the interface to use (normally eth1).
	Local_Port	Denotes the IP port on which the FB will listen.
	Data	The variable used to collect the data at the output, is required as an input, to allow its length to be checked and its "data field" to be located.
Outputs	Size	Denotes the size (in Bytes) of the internally buffered data from a received datagram.
	Data	Each Byte of the internal buffer will be copied to consecutive bytes of the connected variable's "data field." (See remarks).
	Remote_IP_Address	Denotes the IPv4 address from which a datagram has been received.
	Remote_Port	Denotes the source port of a datagram which has been received.
	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	For the data field it is recommended to define a data type such as: <pre> TYPE dataArray_t : ARRAY [0..1024] OF BYTE; END_TYPE </pre> Connect a variable of this type to data in and out. Before stopping the MULTIPROG Express application the FB should be disabled to close the socket.	

Table 44: i102_UDP_ReceiverArray Function Block Description

3.5.4. i102_UDP_SenderArray

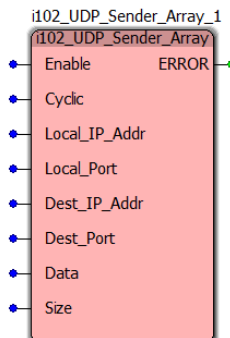
Name	i102_UDP_SenderArray	
Purpose	Used to send an UDP datagram from a sender.	
Module type	i102	
Symbol		
IEC 61131 type	Function block	
Parameters	Enable	TRUE: The sending of datagrams is enabled. FALSE: FB in disabled.
	Cyclic	TRUE: The FB will send datagrams continuously until disabled. FALSE: The FB will send 1 datagram each time the program instance containing the FB is activated.
	Local_IP_addr	Denotes the IPv4 address of the interface to use (normally eth1).
	Local_Port	Denotes the source port of the datagram.
	Dest_IP_Addr	Denotes the destination IPv4 address for the datagram.
	Dest_Port	The port on which the host at the destination address is expected to be listening.
	Data	Can be connected to a complex data type (usually an array of Byte).
	Size	The size in Bytes of the data payload.
Outputs	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	Before stopping the MULTIPROG Express application the FB should be disabled to close the socket.	

Table 45: i102_UDP_SenderArray Function Block Description

3.5.5. ETH_Config

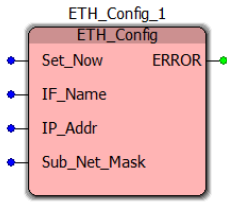
Name	ETH_Config	
Purpose	Configure an Ethernet interface.	
Module type	i102	
Symbol		
IEC 61131 type	Function block	
Parameters	IF_Name	Denotes the name of the interface of which the configuration is to be set. The service interface is normally "eth0" and the fieldbus interface is "eth1".
	IP_Addr	The IPv4 address to be set [e. g. 192.168.1.40].
	Sub_Net_Mask	The IPv4 sub net mask to be set [e. g. 255.255.255.0].
Outputs	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	-	

Table 46: ETH_Config function block description

3.5.6. ETH_Get_Config

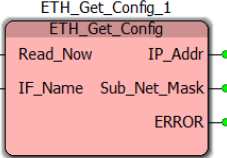
Name	ETH_Get_Config	
Purpose	Receive a string from the caller and confirm the message with a reply string.	
Module type	i101	
Symbol		
IEC 61131 type	Function block	
Parameters	Read_Now	Transition FALSE → TRUE is the trigger point. TRUE: The configuration is read on the rising edge. FALSE: FB is disabled.
	IF_Name	Denotes the name of the interface of which the configuration is to be set. The service interface is normally "eth0" and the fieldbus interface is "eth1".
Outputs	IP_Addr	The IPv4 address to be set [e. g. 192.168.1.40].
	Sub_Net_Mask	The IPv4 sub net mask to be set [e. g. 255.255.255.0].
	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	-	

Table 47: ETH_Get_Config Function Block Description

3.6. i303 Ethernet Module – i303_DG_UDP_LIB Library

3.6.1. i303 Hardware Interface



Figure 15: i303 Ethernet Connectors

Interface	Description
X1	Ethernet Interface 1
X2	Ethernet Interface 2 (optional)

Table 48: i303 Hardware Interface

3.6.2. i303 Software Interface

The following function blocks are included in the i303_DG_UDP_LIB library:

Used Block	Purpose	OP Requirement	i303_UDP_Receiver_Array	i303_UDP_Sender_Array	i303_ETH_Configure	i303_ETH_Get_Config
i303_UDP_Receiver_Array	Receive UDP Datagram	-		-	-	-
i303_UDP_Sender_Array	Send UDP Datagram	-	-		-	-
i303_ETH_Configure	Configure Ethernet Interface	-	-	-		-
i303_ETH_Get_Config	Read Ethernet Configuration	-	-	-	-	

Table 49: i303 Function Blocks Overview

When selecting function blocks for use on the same communication channel please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.6.3. i303_UDP_ReceiverArray

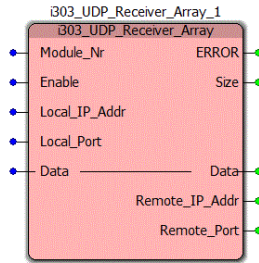
Name	i303_UDP_ReceiverArray	
Purpose	Used to receive an UDP datagram from a sender.	
Module type	i303	
Symbol		
IEC 61131 type	This FB supports only data size of maximum 800Bytes!	
	Function block	
Parameters	Enable	TRUE: Opens a port on the Ethernet interface as defined by its IPv4 address and begins to listen for UDP datagrams. FALSE: FB in disabled.
	Local_IP_addr	Denotes the IPv4 address of the interface to use (normally eth1).
	Local_Port	Denotes the IP port on which the FB will listen.
	Data	The variable used to collect the data at the output, is required as an input, to allow its length to be checked and its "data field" to be located.
Outputs	Size	Denotes the size (in Bytes) of the internally buffered data from a received datagram.
	Data	Each Byte of the internal buffer will be copied to consecutive bytes of the connected variable's "data field." (See remarks).
	Remote_IP_Address	Denotes the IPv4 address from which a datagram has been received.
	Remote_Port	Denotes the source port of a datagram which has been received.
	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	For the data field it is recommended to define a data type such as: <pre> TYPE dataArray_t : ARRAY [0..1024] OF BYTE; END_TYPE </pre> Connect a variable of this type to data in and out. Before stopping the MULTIPROG Express application the FB should be disabled to close the socket.	

Table 50: i303_UDP_ReceiverArray Function Block Description

3.6.4. i303_UDP_SenderArray

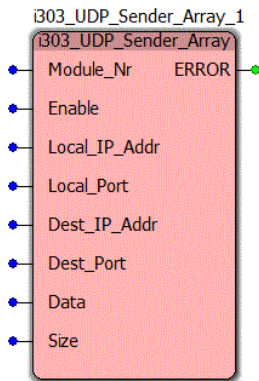
Name	i303_UDP_SenderArray	
Purpose	Used to send an UDP datagram from a sender.	
Module type	i303	
Symbol		
IEC 61131 type	Function block	
	This FB supports only data size of maximum 800Bytes!	
Parameters	Enable	TRUE: The sending of datagrams is enabled. FALSE: FB in disabled.
	Cyclic	TRUE: The FB will send datagrams continuously until disabled. FALSE: The FB will send 1 datagram each time the program instance containing the FB is activated.
	Local_IP_addr	Denotes the IPv4 address of the interface to use (normally eth1).
	Local_Port	Denotes the source port of the datagram.
	Dest_IP_Addr	Denotes the destination IPv4 address for the datagram.
	Dest_Port	The port on which the host at the destination address is expected to be listening.
	Data	Can be connected to a complex data type (usually an array of Byte).
	Size	The size in Bytes of the data payload.
	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	Before stopping the MULTIPROG Express application the FB should be disabled to close the socket.	

Table 51: i303_UDP_SenderArray Function Block Description

3.6.5. i303_ETH_Config

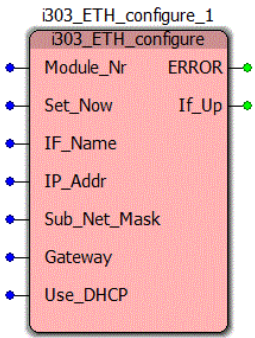
Name	ETH_Configure	
Purpose	Configure an Ethernet interface.	
Module type	i303	
Symbol		
IEC 61131 type	Function block	
Parameters	IF_Name	Denotes the name of the interface of which the configuration is to be set. The service interface is normally "eth0" and the fieldbus interface is "eth1".
	IF_Name	Denotes the name of the interface to be configured [ETH0, ETH1, eth0 or eth1].
	IP_Addr	The IPv4 address to be set [e. g. 192.168.1.40].
	Sub_Net_Mask	The IPv4 sub net mask to be set [e. g. 255.255.255.0].
	Gateway	The IPv4 gateway address to be set [e. g. 192.168.1.1].
	Use_DHCP	Set TRUE to cause the interface to use DHCP to get all settings automatically.
Outputs	ERROR	A non-zero value appearing here indicates that an error has occurred.
	If_Up	TRUE: Link is up and a cable is connected. FALSE: Link is down.
Remarks	-	

Table 52: i303_ETH_Configure function block description

3.6.6. i303_ETH_Get_Config

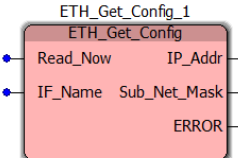
Name	ETH_Get_Config	
Purpose	Receive a string from the caller and confirm the message with a reply string.	
Module type	i101	
Symbol		
IEC 61131 type	Function block	
Parameters	Read_Now	Transition FALSE → TRUE is the trigger point. TRUE: The configuration is read on the rising edge. FALSE: FB is disabled.
	IF_Name	Denotes the name of the interface of which the configuration is to be set. The service interface is normally "eth0" and the fieldbus interface is "eth1".
	IP_Addr	The IPv4 address as read at the Read_Now trigger point.
	Sub_Net_Mask	The IPv4 sub net mask as read at the Read_Now trigger point.
	Gateway	The IPv4 gateway address as read at the Read_Now trigger point.
	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	-	

Table 53: ETH_Get_Config Function Block Description

3.7. Common ionia™ Function Blocks Library – ionia_DG_FW_LIB

3.7.1. ionia™ Software Interface

The following function blocks are included in the ionia_DG_FW_LIB library:

Function	Purpose	Type	PR	MA	RE
SetTimeOfDay	Set the time.	control			
GetTimeOfDay	Get the current time.	control			
Reboot	Restart the PLC	control			
Watch_Dog	Restart the PLC if the application stops	control			
Record_To_File	Record Process Data to the ionia™ storage	control			

Table 54: Common ionia™ Function Blocks Overview

When selecting function blocks for use on the same central processing unit module please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.7.2. SetTimeOfDay

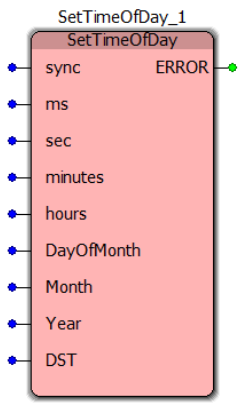
Name	SetTimeOfDay	
Purpose	To set the (system) time of day.	
Module type	ionia™	
Symbol		
IEC 61131 type	Function block	
Parameters	sync	FALSE → TRUE the time of day is transferred to the ionia™ PLC. TRUE: On the rising edge FALSE: FB in disabled.
	ms	The number of milliseconds after the second.
	sec	The number of seconds after the minute.
	minutes	The number of minutes after the hour.
	hours	The number of hours since midnight.
	DayOfMonth	Day of month [1..31].
	Month	Calendar month [1..12].
	Year	Calendar year.
	DST	Positive value: Daylight saving in effect. Zero: Daylight saving not in effect. Negative value: Information is not available.
Outputs	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	-	

Table 55: SetTimeOfDay Function Block Description

3.7.3. GetTimeOfDay

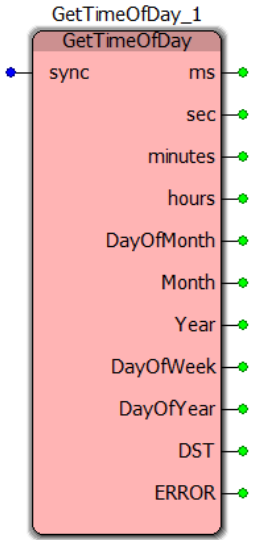
Name	GetTimeOfDay	
Purpose	To read the (system) time of day.	
Module type	ionia™	
Symbol		
IEC 61131 type	Function block	
Parameters	sync	FALSE → TRUE the time of day is transferred to the ionia™ PLC. TRUE: On the rising edge FALSE: FB in disabled.
Outputs	ms	The number of milliseconds after the second.
	sec	The number of seconds after the minute.
	minutes	The number of minutes after the hour.
	hours	The number of hours since midnight.
	DayOfMonth	Day of month [1..31].
	Month	Calendar month [1..12].
	Year	Calendar year.
	DayOfWeek	Days since Sunday [0..6].
	DayOfYear	Days since 1st of January [0..365].
	DST	Positive value: Daylight saving in effect. Zero: Daylight saving not in effect. Negative value: Information is not available.
	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	-	

Table 56: GetTimeOfDay Function Block Description

3.7.4. Reboot

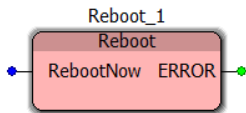
Name	Reboot	
Purpose	To restart the PLC (reboots the ionia™ PLC's CPU).	
Module type	ionia™	
Symbol		
IEC 61131 type	Function block	
Parameters	RebootNow	FALSE → TRUE the ionia™ PLC will be rebooted. TRUE: Reboot ionia™ PLC FALSE: FB in disabled.
Outputs	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	-	

Table 57: Reboot Function Block Description

3.7.5. Watch_Dog

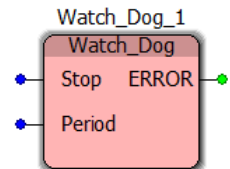
Name	Watch_Dog	
Purpose	To restart the PLC (reboots the ionia™ PLC's CPU) if the application stops.	
Module type	ionia™	
Symbol		
IEC 61131 type	Function block	
Parameters	RebootNow	FALSE: The watchdog runs. If the application stops the watchdog will reboot the ionia™ PLC. TRUE: The watchdog is stopped. The application can be safely stopped.
	Period	Sets the watchdog period. If the POU containing the running Watch_Dog FB is not called within this period the PLC will be rebooted. Input value 0 (or not connected): the default period of 100ms is used Input value 1... 511: a period of 1 to 511 ms is used Out of range values result in the default period and ERROR=596 (parameter exceeds bounds)
Outputs	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	This FB is particularly useful in a redundant ionia™ configuration.	

Table 58: Watch_Dog Function Block Description

3.7.6. Record_To_File

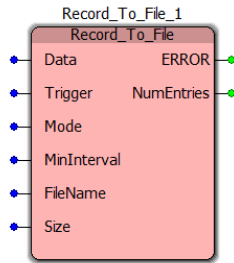
Name	Record_To_File	
Purpose	Records process data to a file on the ionia™ filesystem.	
Module type	ionia™	
Symbol		
IEC 61131 type	Function block	
Parameters	Data	Current value of data to be recorded.
	Trigger	TRUE: Record current value to entry in file FALSE: Do nothing
	Mode	0: Record to pre-allocated ring buffer file 1: Record to file with max. size
	MinInterval	If no trigger was set for this time, then record the current value.
	FileName	Name of the file for later log retrieval.
	Size	Maximum size of the logfile.
	NumEntries	Current number of entries in the logfile.
Outputs	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	The data can be any of datatype. <i>struct</i> datatypes are recorded as is, the user has to retrieve the individual signals from the logfile. Each entry is timestamped with the current system time in UTC.	

Table 59: Record_To_File Function Block Description

3.8. Redundant Operation – i101_redundancy_LIB (optional)

This library is optional and not included in the standard duagon library set. The order code for this library is i402. Please contact duagon for more information.

3.8.1. Redundant Operation Software Interface

The following function blocks are included in the i101_redundancy_LIB library:

Function	Purpose	Type	PR	MA	RE
i101_App_Supervision	To control the operation of two PLCs in a redundant configuration.	control			
i101_VCUVCU_PORT	To provide information to the application from the remote instance of two PLCs in a redundant configuration.	control			

Table 60: Redundant Operation Function Blocks Overview

When selecting function blocks for use on the same central processing unit module please consider the combination strategy indicated by the last columns and the following key:

- PR Prohibited
- MA Mandatory
- RE Recommended

3.8.2. Principle of Operation

A redundant system is given by two identical ionia™ PLCs, running the same MULTIPROG Express application. At any time only one of the PLCs is actively involved in the control process: the ACTIVE device. The other is able to passively monitor the process: the PASSIVE device.

Which of the two controllers becomes active at start-up is decided by designating one as SLAVE. The other is then per default MASTER. During operation the MASTER/SLAVE role does not change. The designation of the SLAVE controller is done by use of the Web front-end as described in section “Designating the SLAVE controller” below.

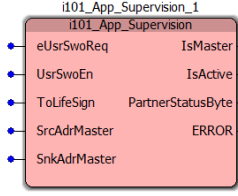
When starting, the MASTER checks the SLAVE. If the SLAVE is already active it does not become active. Otherwise, if the SLAVE is passive, it becomes active straight away. The slave on the other hand starts-up in passive mode and then waits a short period, designed to compensate for any differences in starting time. Then it begins to check the activity of the MASTER controller.

To use an ionia™ PLC in a redundant configuration a special FB “i101_App_Supervision” must be included in the MULTIPROG project. This allows the designation of ports for monitoring the partner controller (one in each direction). It also provides a means of initiating or inhibiting a switchover and communicating status information between the partner controllers' applications.

To provide for further communication requirements between the partners the “i101_VCUVCU_PORT” FB is provided.

Because ports associated with these FBs are different in the two controllers (SOURCE at opposite ends) there are special MVB configuration considerations as explained in section “Configuration for Redundancy” below.

3.8.3. i101_App_Supervision

Name	i101_App_Supervision	
Purpose	To control the operation of two PLCs in a redundant configuration.	
Module type	i101_Redundancy	
Symbol		
IEC 61131 type	Function block	
Parameters	eUsrSwoReq	FALSE: No user switch-over request. TRUE: On the rising edge FALSE → TRUE a user switch-over sequence is requested.
	UsrSwoEn	FALSE: disables eUsrSwoReq. TRUE: enables eUsrSwoReq.
	ToLifeSign	Sets the time in seconds after which a non-changing life-sign count will be detected. Range: t#0.001s ... t#60s. Resolution 0.001s. After expiry of this time-out at the PASSIVE instance, freshness is monitored on the SINK port. Then, if the data becomes stale, the PASSIVE instance becomes ACTIVE because this means the partner controller (which was previously ACTIVE) is no longer active on the MVB.
	SrcAdrMaster	One of two MVB port addresses used for communication between the two PLCs in a redundant configuration. The "Master" will configure this as a SOURCE port. The "Slave" will configure this as a SINK port.
	SnkAdrMaster	The second of two MVB port address used for communication between the two PLCs in a redundant configuration. The "Master" will configure this as a SINK port. The "Slave" will configure this as a SOURCE port.
Outputs	IsMaster	TRUE: this ionia™ PLC is "Master". FALSE: this ionia™ PLC is "Slave". At system start-up the "Master" instance will become ACTIVE and the "Slave" PASSIVE.
	IsActive	TRUE: this ionia™ PLC is ACTIVE. FALSE: this ionia™ PLC is PASSIVE. The PASSIVE device has disabled all SOURCE ports apart from those used for dedicated communication between the redundant partners.
	PartnerStatusByte	This byte contains a bit-map in accordance with the table below.

Remarks	ERROR	A non-zero value appearing here indicates that an error has occurred.
	Please note that the "SrcAdrMaster" and the "SnkAdrMaster" addresses have to be consistent with the MVB configuration of the MASTER and SLAVE devices.	

Table 61: i101_App_Supervision Function Block Description

PartnerStatusByte

Bit	Description
0	1: Partner has user eUsrSwoReq set TRUE. 0: Partner has user eUsrSwoReq set FALSE.
1	1: Partner has user UsrSwoEn set TRUE. 0: Partner has user UsrSwoEn set FALSE.
2	1: Partner is "Master". 0: Partner is "Slave".
3	1: Partner is ACTIVE. 0: Partner is PASSIVE.
4	RESERVED. At present undefined and should be ignored as it might be allocated in future.
5	RESERVED. At present undefined and should be ignored as it might be allocated in future.
6	RESERVED. At present undefined and should be ignored as it might be allocated in future.
7	RESERVED. At present undefined and should be ignored as it might be allocated in future.

Table 62: PartnerStatusByte Description

3.8.4. i101_VCUVCU_PORT

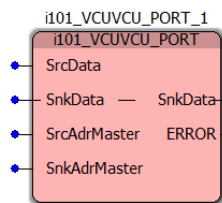
Name	i101_VCUVCU_PORT	
Purpose	To provide information to the application from the remote instance of two PLCs in a redundant configuration.	
Module type	i101_Redundancy	
Symbol		
IEC 61131 type	Function block	
Parameters	SrcData	Data to be sent to the remote PLC.
	SrcAdrMaster	One of two MVB port address used for communication between the i101_VCUVCU_PORT function blocks on each of two PLCs in a redundant configuration. The "Master" will configure this as a SOURCE port. The "Slave" will configure this as a SINK port.
	SnkAdrMaster	The second of two MVB port address used for communication between the i101_VCUVCU_PORT function blocks on each of two PLCs in a redundant configuration. The "Master" will configure this as a SINK port. The "Slave" will configure this as a SOURCE port.
	SnkData	The variable used to provide the data at the output, is required as an input, to allow its length to be checked and its "data field" to be located.
Outputs	SnkData	Data received from the remote PLC.
	ERROR	A non-zero value appearing here indicates that an error has occurred.
Remarks	The SOURCE port remains active on the passive ionia™ PLC allowing the exchange of data to continue. Please note that the "SrcAdrMaster" and the "SnkAdrMaster" addresses have to be consistent with the MVB configuration of the MASTER and SLAVE devices.	

Table 63: i101_VCUVCU_PORT Function Block Description

4. duagon I/O Driver

This chapter gives information regarding the duagon IO driver. IO driver provide, in contrast to the FW Libraries, communication by means of fieldbus systems. The name IO Driver is therefore to understand as fieldbus input/output communication, whereas the firmware library is mainly used to drive/acquire the ionia™ periphery input/output channels.

In this section all steps needed to accomplish field bus communication are explained from a technical point of view, for each single step. The whole work flow is described in "Getting Started with ionia™ – User's Guide" (d-007724-nnnnnn), this document has to be referenced for questions regarding details on the process of building fieldbus communication systems.

4.1. MVB PD Communication - i101/i301_DG_MVB_DRV

The i101_DG_MVB_DRV is for MVB process data communication. Various steps must be processed in order to establish process data communication over MVB. These steps are:

1. Write a MVB configuration file, i.e. a D2000 file.
2. In MULTIPROG Express:
 - a. Add all IO Groups corresponding to the MVB Ports specified in the configuration file.
 - b. Add all process variables with its port address and byte offset to the *Global Variable Worksheet*.

The IO group configuration as well as the definition in the *Global Variables Worksheet* are done from the *Hardware* tab of the *Project Wizard* of MULTIPROG Express.

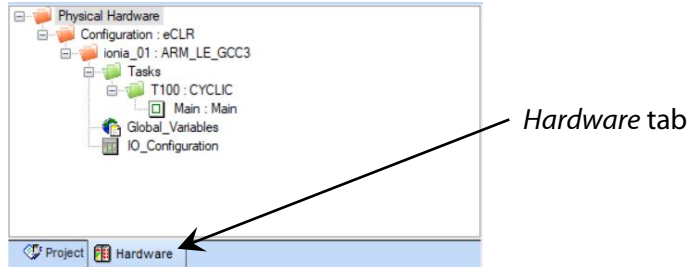


Figure 16: MULTIPROG Express *Hardware* tab (MVB PD Communication)

4.1.1. MVB Configuration File

The syntax and handling of the D2000 tool is described in "D2000 Configuration Tool - Data Sheet", d-000341-nnnnnn. Only a short excerpt of a D2000 configuration file is shown below:

Device specification section

```

/*****
/* The following section defines the devices list.          */
/*****
$$DEVICES

$$$DEVICE   ionia_VCU_01   0x10   BAFULL
$$$MD_PARAMETERS
$$$$$MAX_CALL_NUMBER      5
$$$$$MAX_INST_NUMBER      10
$$$$$DEFAULT_REPLY_TIMEOUT 400
$$$$$MY_CREDIT            7

$$$DEVICE
$$$DEVICE   i101_BRK_01   0x70   BAFULL
$$$MD_PARAMETERS
$$$$$MAX_CALL_NUMBER      5
$$$$$MAX_INST_NUMBER      10
$$$$$DEFAULT_REPLY_TIMEOUT 400
$$$$$MY_CREDIT            7

$$$DEVICE
$$$DEVICE   i101_DCU_01   0x80   BAFULL
$$$MD_PARAMETERS
$$$$$MAX_CALL_NUMBER      5
$$$$$MAX_INST_NUMBER      10
$$$$$DEFAULT_REPLY_TIMEOUT 400
$$$$$MY_CREDIT            7

.....

```

Process data telegram section

```

/*****
/* The following section defines the process data telegrams */
/* between intelligent devices.                             */
/*****
$$PD_COMM
$$$PD_TELEGRAM   Tgm_BRK_SS_01   i101_VCU_01   0x600   B8   32   AUTO
$$$$$SINK_DEVICE i101_BRK_01
$$$$$SINK_DEVICE i101_DCU_01

$$$PD_TELEGRAM   Tgm_DCU_SS_01   i101_VCU_01   0x700   B8   32   AUTO
$$$$$SINK_DEVICE i101_BRK_01
$$$$$SINK_DEVICE i101_DCU_01

.....

```

For all MVB bus participants a device must be specified as well as the MVB process data communication between the devices.

The telegram numbers correspond with the input and output ports of a certain device.

Parsing the MVB Configuration File

The MVB configuration file is parsed by means of the D2000 tool according to "D2000 Configuration Tool - Data Sheet". The output of the parse process is among other a bin file for each defined device. The above excerpt of the device configuration generates the following bin files:

- i101_VCU_01.bin
- i101_BRK_01.bin
- i101_DCU_01.bin

Download the MVB Configuration

The appropriate bin file is downloaded to the device. This may be done by means of MULTIPROG Express according to "Getting Started with ionia™ – User's Guide", or by means of the ionia™ web front end.

Downloading the MVB configuration via MULTIPROG Express requires two additional steps:

1. Put the bin File in your MULTIPROG Express project directory (*MPE_PROJDIR*):
`%MPE_PROJDIR%\C\Configuration\R\ionia_01\`
2. Add the bin file to the eCLRDeviceProfile.xml located in the MULTIPROG Express installation directory (*MPE_INSTALLDIR*):
`%MPE_INSTALLDIR%\plc\eCLR\`

A detailed description of all these steps can be found in "Getting Started with ionia™ – User's Guide", d-007724-nnnnnn. The configuration bin-file will be packed into the boot image file of the project and automatically unpacked at run time to the correct directory.

Download the MVB configuration via web front-end requires a tar file which copies the MVB configuration to the following path on the device:

`/home/root/duagon/application/`

Note: An MVB configuration file downloaded with the web front-end will be overwritten at run-time if the MULTIPROG Express boot image file contains a configuration file.

In both cases, it is crucial that the device name resp. the BIN FILE NAME AND the RESOURCE NAME of MULTIPROG Express are identical.

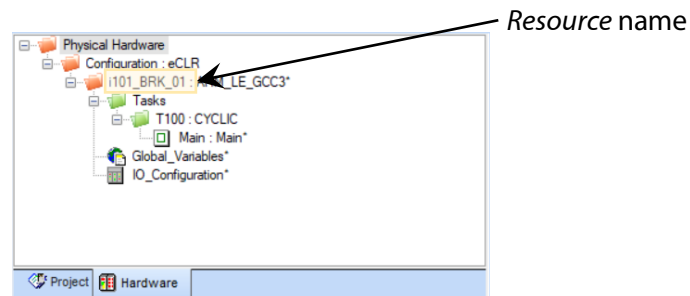


Figure 17: MULTIPROG Express Resource Name

Download Procedure

There is more than one possibility to download the MVB configuration: the i101_DG_MVB_DRV decides on how to configure the duagon MVB controller (DMC) by employing the following algorithm (see Figure 18):

- If the application image contains no configuration file, the i101_DG_MVB_DRV driver just checks if a configuration file is on the flash.
- The configuration file stored on the flash might be from a previous download or downloaded individually.
- If no configuration is found, an error is emitted in MULTIPROG Express.

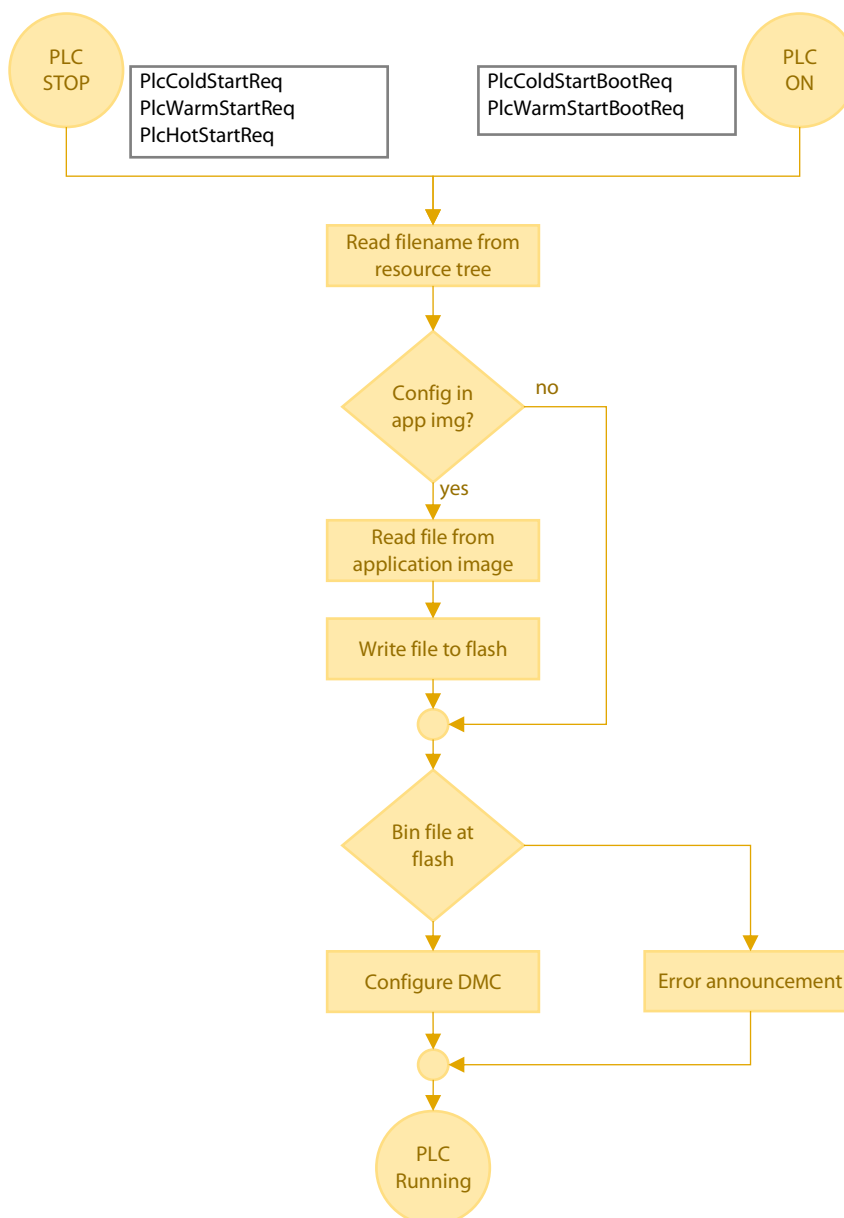


Figure 18: Configuration Loading Algorithm

MULTIPROG Express IO Configuration

Since the duagon MVB controller is configured by means of the D2000 configuration file at run time, a mechanism is needed to tell the IEC 61131 MULTIPROG Express application which groups exist in the project, this is the job of the IO groups.

The input and output telegrams are represented by a corresponding IO group in MULTIPROG Express. The IO group start address is calculated from the port address using the following formula:

- MVB Port * 1000

This start address has to be entered into the property sheet of the IO configuration.

Figure 19 shows an example configuration for the MVB port 0x100. The ports are read respectively written as one data chunk.

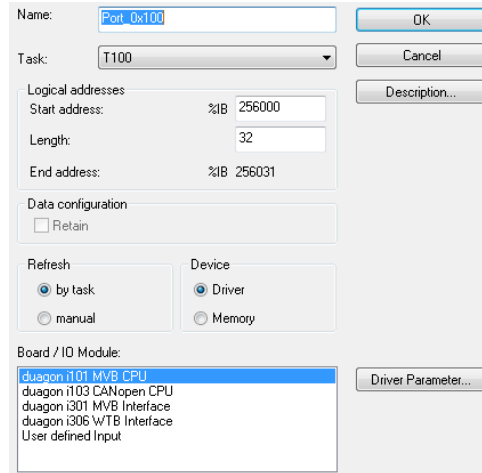


Figure 19: IO Group Configuration Example

4.1.2. Configuration for Redundancy

Terminology

The following terms are introduced in the context of a redundant ionia™ configuration:

- MASTER / SLAVE: During commissioning, one ionia™ PLC is designated to be the MASTER device, while the other ionia™ PLC is designated as the SLAVE device. Only one device can be designated as the MASTER and only one can be the SLAVE. During the operation this designation remains constant.
- ACTIVE / PASSIVE: During normal operation one ionia™ PLC is the ACTIVE device and the second one is PASSIVE. The ACTIVE device is in charge of performing the process control tasks, while the PASSIVE one is in “stand-by” and monitors the activity of its partner. The PASSIVE device is ready to take over control in case the ACTIVE one fails.

Introduction

In a redundant configuration the MVB SOURCE ports requiring enhanced availability will be sourced from either of two PLCs. To this end, the two PLCs have (typically) identical MULTIPROG projects containing an i101_App_Supervision FB.

They also each have two MVB configuration files containing identical specifications for the redundant ports. The configuration files also have to specify two ports used as supervision ports by the i101_App_Supervision FB. These two ports are used to exchange state information between the two PLCs and to coordinate switch-over. When required, the MVB configuration also defines a number of SOURCE/SINK port pairs used for data communication between the two PLCs by means of "i101_VCUVCU_PORT" FBs.

The two MVB configuration files are named:

<RESOURCE NAME>.bin (typically: ionia_01.bin)

and

<RESOURCE NAME>_s.bin (typically ionia_01_s.bin)

Both files are downloaded onto each PLC. During the initialization phase, the MASTER device will use the <RESOURCE NAME>.bin configuration file, whereas the SLAVE device the <RESOURCE NAME>_s.bin configuration file.

Start-up

The SLAVE:

1. Disables SOURCE ports.
2. Waits for a short delay before performing the next initialization steps.
3. The MVB IO driver configures the ports according to its .bin configuration file.
4. The i101_App_Supervision FB designates the SnkAdrMaster port, which is configured as SOURCE in the SLAVE, as "always active" (even if the device is in its PASSIVE state).
5. Becomes PASSIVE.

The MASTER:

1. MVB IO driver configures the ports according to its bin.
2. The i101_App_Supervision FB designates the SrcAdrMaster port as "always active".
3. Checks if there is already some activity on the MVB.
4. If no MVB activity from the SLAVE is detected it becomes ACTIVE (enabling the SOURCE ports). Otherwise, if the SLAVE is already ACTIVE, it becomes PASSIVE (disabling the SOURCE).

Designating the SLAVE Controller

To designate one controller as SLAVE using the web front-end, click the *Set Redundancy Slave* button on the *IEC Interface* side tab of the main *i101* tab as shown below. The button will change to *Unset Redundancy Slave*.

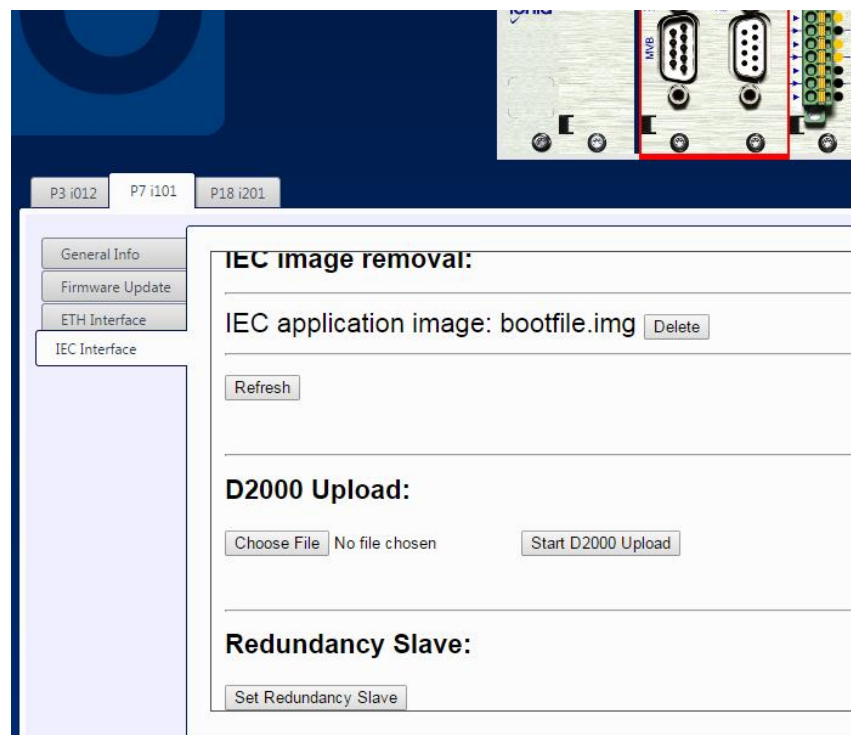


Figure 20: PLC Web Front-end

The designation as SLAVE works by creating a special file on the ionia™ PLC. If this ionia™ PLC is to be used later in another configuration, the file will have to be removed. This can be done by use of the *Unset Redundancy Slave* button. Normally this is the only reason to use this button as the designation of MASTER and SLAVE does not ordinarily change in an installed system.

Creation of .bin Files Using D2000 tool

On each PLC the local device name is (typically) ionia_01 and the remote device is named ionia_01_slave or ionia_01_master respectively. So that the entry in the \$\$\$DEVICES section of a typical D2000 configuration file for the MASTER looks like this:

```
$$$DEVICE    ionia_01          0x10    BAFULL
.....
$$$DEVICE    ionia_01_slave    0x11    BAFULL
```

And for the SLAVE like this:

```
$$$DEVICE    ionia_01_master    0x10    BAFULL
.....
$$$DEVICE    ionia_01_slave    0x11    BAFULL
```

The ports used to communicate between the i101_App_Supervision FBs are specified differently in the two configuration files. The corresponding section of a typical D2000 configuration file for the MASTER looks like this:

```
/* **** */
/* The following section defines the supervision data */
/* telegrams between master and slave. */
/* **** */
$$$PD_TELEGRAM    Tgm_1024    ionia_01_slave    0x400    B32    32    AUTO
$$$SINK_DEVICE    ionia_01

$$$PD_TELEGRAM    Tgm_1056    ionia_01          0x420    B32    32    AUTO
$$$SINK_DEVICE    ionia_01_slave
.....
```

And for the SLAVE like this:

```
/* **** */
/* The following section defines the supervision data */
/* telegrams between master and slave. */
/* **** */
$$$PD_TELEGRAM    Tgm_1024    ionia_01          0x400    B32    32    AUTO
$$$SINK_DEVICE    ionia_01

$$$PD_TELEGRAM    Tgm_1056    ionia_01_master    0x420    B32    32    AUTO
$$$SINK_DEVICE    ionia_01_slave
.....
```

With the following directory structure:

```
current_directory\indata\
current_directory\master\
current_directory\slave\
```

and these D2000 configuration files based on the above description:

```
current_directory\indata\master.txt
current_directory\indata\slave.txt
```

and d2000.exe in the current_directory\ or in the PATH environment variable; the command lines:

```
d2000 indata master.txt master
d2000 indata slave.txt slave
```

will result in the following configuration files, which are required:

```
current_directory\master\ionia_01.bin
current_directory\slave\ionia_01.bin
```

Configuration files ionia_01_slave.bin and ionia_01_master.bin are not required. Further configuration files such as D412.bin can be taken from the master directory.

VCU-VCU Communication

To allow the two PLCs involved in the redundant configuration to communicate with each-other, the i101_VCUVCU_PORT FB is provided. This FB gives the application continuous access to data from the ACTIVE PLC whether that be the local or remote controller. For each such block used in the project, a D2000 configuration is required.

The corresponding section of a typical D2000 configuration file for the MASTER looks like this:

```

/*****
/* The following section defines the supervision data */
/* telegrams between master and slave. */
/*****
$$$$PD_TELEGRAM      Tgm_1088  ionia_01_slave  0x440 B32 32  AUTO
$$$$SINK_DEVICE      ionia_01

$$$$PD_TELEGRAM      Tgm_1120  ionia_01        0x460 B32 32  AUTO
$$$$SINK_DEVICE      ionia_01_slave
.....

```

And for the Slave it looks like this:

```

/*****
/* The following section defines the supervision data */
/* telegrams between master and slave. */
/*****
$$$$PD_TELEGRAM      Tgm_1088  ionia_01        0x440 B32 32  AUTO
$$$$SINK_DEVICE      ionia_01_master

$$$$PD_TELEGRAM      Tgm_1120  ionia_01_master 0x460 B32 32  AUTO
$$$$SINK_DEVICE      ionia_01
.....

```

Example

The MULTIPROG Express project i101_MVB_GetStarted_RED simple example of a project for a pair of PLCs to be used in a redundant configuration. The D2000 configuration file for the slave looks like this:

```

/*****
/*****
/**          MVB Description          **
/*****
/*****

/*****
/* The following section describes the project. */
/*****
$PROJECT      i101_MS  1.0.0    2014-10-24

/*****
/* The following section defines the bus MVB 0. */
/* All sections below are part of MVB 0. */
/*****
$MVB      0      LINE_BOTH

/*****
/* The following section defines the devices list. */
/*****
$DEVICES

$$$$DEVICE      ionia_01_master 0x10    BAFULL
$$$$MD_PARAMETERS
$$$$$MAX_CALL_NUMBER      5
$$$$$MAX_INST_NUMBER      10
$$$$$DEFAULT_REPLY_TIMEOUT 400
$$$$$MY_CREDIT      7

```

```

$$$$DEVICE ionia_01          0x11  BAFULL
$$$$SMD_PARAMETERS
$$$$$MAX_CALL_NUMBER        5
$$$$$MAX_INST_NUMBER        10
$$$$$DEFAULT_REPLY_TIMEOUT  400
$$$$$MY_CREDIT              7

$$$$DEVICE d412             0x20  BAFULL
$$$$SMD_PARAMETERS
$$$$$MAX_CALL_NUMBER        5
$$$$$MAX_INST_NUMBER        10
$$$$$DEFAULT_REPLY_TIMEOUT  400
$$$$$MY_CREDIT              7

/*****
/* The following section defines the Bus Administrator(s). */
/*****
$$BA

// List of devices that act as bus administrators.
$$BA_DEVICE          ionia_01

// Parameter definition, valid for all BA devices on this MVB.
$$BA_PARAMETERS
$$$$ACTUALISATION_KEY      0x0000
$$$$T_REPLY_MAX           0
$$$$EVENT_POLL_STRATEGY   LOW
$$$$BASIC_PERIOD          1000
$$$$MACROCYCLES_PER_TURN  16
$$$$DEVICES_SCAN_STRATEGY ALL

/*****
/* The following section defines the process data telegrams */
/* between intelligent devices. */
/*****
$$PD_COMM

$$$$PD_TELEGRAM      Tgm_256 d412          0x100 B32 32  AUTO
$$$$$SINK_DEVICE     ionia_01

$$$$PD_TELEGRAM      Tgm_512 ionia_01      0x200 B32 32  AUTO
$$$$$SINK_DEVICE     d412

/*****
/* The following section defines the supervision data */
/* telegrams between master and slave. */
/*****
$$$$PD_TELEGRAM      Tgm_1024 ionia_01      0x400 B32 32  AUTO
$$$$$SINK_DEVICE     ionia_01_master

$$$$PD_TELEGRAM      Tgm_1056 ionia_01_master 0x420 B32 32  AUTO
$$$$$SINK_DEVICE     ionia_01

/*****
/* The following section defines the VCU VCU data telegrams */
/* between master and slave. */
/*****
$$$$PD_TELEGRAM      Tgm_1088 ionia_01      0x440 B32 32  AUTO
$$$$$SINK_DEVICE     ionia_01_master

$$$$PD_TELEGRAM      Tgm_1120 ionia_01_master 0x460 B32 32  AUTO
$$$$$SINK_DEVICE     ionia_01

/*****
/* The following section defines the process data telegrams */
/* between intelligent and IO-devices. */
/*****
$$PD_IO

```

The D2000 configuration for the MASTER looks like this:

```

/*****
/*****
/**          MVB Description          **/
/*****
/*****

/*****
/* The following section describes the project.          */
/*****
$PROJECT    i101_MS    1.0.0    2014-10-24

/*****
/* The following section defines the bus MVB 0.          */
/* All sections below are part of MVB 0.                  */
/*****
$MVB    0    LINE_BOTH

/*****
/* The following section defines the devices list.        */
/*****
$$DEVICES

$$$$DEVICE    ionia_01          0x10    BAFULL
$$$$SMD_PARAMETERS
$$$$$MAX_CALL_NUMBER          5
$$$$$MAX_INST_NUMBER          10
$$$$$DEFAULT_REPLY_TIMEOUT    400
$$$$$MY_CREDIT                7

$$$$DEVICE    ionia_01_slave    0x11    BAFULL
$$$$SMD_PARAMETERS
$$$$$MAX_CALL_NUMBER          5
$$$$$MAX_INST_NUMBER          10
$$$$$DEFAULT_REPLY_TIMEOUT    400
$$$$$MY_CREDIT                7

$$$$DEVICE    d412              0x20    BAFULL
$$$$SMD_PARAMETERS
$$$$$MAX_CALL_NUMBER          5
$$$$$MAX_INST_NUMBER          10
$$$$$DEFAULT_REPLY_TIMEOUT    400
$$$$$MY_CREDIT                7

/*****
/* The following section defines the Bus Administrator(s). */
/*****
$$BA

// List of devices that act as bus administrators.
$$BA_DEVICE          ionia_01

// Parameter definition, valid for all BA devices on this MVB.
$$BA_PARAMETERS
$$$$$ACTUALISATION_KEY        0x0000
$$$$$T_REPLY_MAX              0
$$$$$EVENT_POLL_STRATEGY      LOW
$$$$$BASIC_PERIOD              1000
$$$$$MACROCYCLES_PER_TURN     16
$$$$$DEVICES_SCAN_STRATEGY    ALL

/*****
/* The following section defines the process data telegrams */
/* between intelligent devices.                               */
/*****
$PD_COMM

```

```

$$$PD_TELEGRAM      Tgm_256  d412                0x100 B32 32  AUTO
$$$$$SINK_DEVICE    ionia_01

$$$PD_TELEGRAM      Tgm_512  ionia_01            0x200 B32 32  AUTO
$$$$$SINK_DEVICE    d412

/*****/
/* The following section defines the supervision data */
/* telegrams between master and slave. */
/*****/
$$$PD_TELEGRAM      Tgm_1024  ionia_01_slave 0x400 B32 32  AUTO
$$$$$SINK_DEVICE    ionia_01

$$$PD_TELEGRAM      Tgm_1056  ionia_01          0x420 B32 32  AUTO
$$$$$SINK_DEVICE    ionia_01_slave

/*****/
/* The following section defines the VCU VCU data telegrams */
/* between master and slave. */
/*****/
$$$PD_TELEGRAM      Tgm_1088  ionia_01_slave 0x440 B32 32  AUTO
$$$$$SINK_DEVICE    ionia_01

$$$PD_TELEGRAM      Tgm_1120  ionia_01          0x460 B32 32  AUTO
$$$$$SINK_DEVICE    ionia_01_slave

/*****/
/* The following section defines the process data telegrams */
/* between intelligent and IO-devices. */
/*****/
$$$PD_IO

```

4.1.3. MULTIPROG Express Global Variables Worksheet

The process variables themselves are accessed by calling an address as shown below:

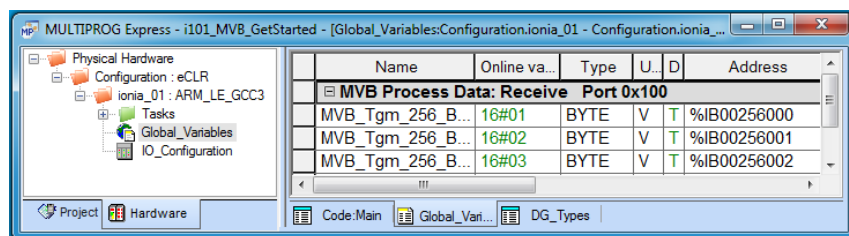
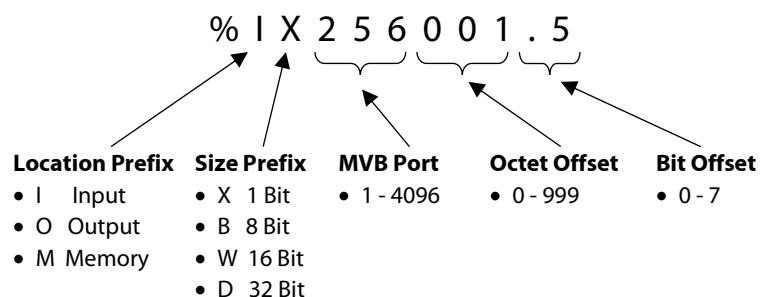


Figure 21: Global Variables Declarations

The address itself is composed according to the following scheme:



The above example points to the 6th Bit of the second Byte of port 0x100. More information regarding the process of configuring the IO driver can be found in "Getting Started with ionia™ – User's Guide", d-007724-nnnnnn.

4.1.4. Specific Global MVB IO Addresses

Specific MVB controller and/or driver status information can be read out through accessing the address of the shared memory as listed below.

Name	Type	Description	Address
MVB_INITIALIZED	BOOL	MVB Controller is initialized	%MX 3.1000000.0

Table 64: Initialized Status

Name	Type	Description	Address
MVB_DEVICE_ADDRESS	WORD	MVB device address	%MX 3.1000003
DEVICE_STATUS_WORD	WORD	DSW, 0xFFFF DSW not valid	%MX 3.1000001
DEVICE_STATUS_BIT_SP	BOOL	TRUE: Special device	%MX 3.1000002.7
DEVICE_STATUS_BIT_BA	BOOL	TRUE: Bus administrator capability	%MX 3.1000002.6
DEVICE_STATUS_BIT_GW	BOOL	TRUE: Groupway capability	%MX 3.1000002.5
DEVICE_STATUS_BIT_MD	BOOL	TRUE: Message data capability	%MX 3.1000002.4
DEVICE_STATUS_BIT_CS3	BOOL	Device specific field bit 3	%MX 3.1000002.3
DEVICE_STATUS_BIT_CS2	BOOL	Device specific field bit 2	%MX 3.1000002.2
DEVICE_STATUS_BIT_CS1	BOOL	Device specific field bit 1	%MX 3.1000002.1
DEVICE_STATUS_BIT_CS0	BOOL	Device specific field bit 0	%MX 3.1000002.0
DEVICE_STATUS_BIT_LAT	BOOL	TRUE: Line A trusted	%MX 3.1000001.7
DEVICE_STATUS_BIT_RLD	BOOL	TRUE: Redundant line disturbed	%MX 3.1000001.6
DEVICE_STATUS_BIT_SSD	BOOL	TRUE: Some system disturbance	%MX 3.1000001.5
DEVICE_STATUS_BIT_SDD	BOOL	TRUE: Some device disturbance	%MX 3.1000001.4
DEVICE_STATUS_BIT_ERD	BOOL	TRUE: Extended reply time-out	%MX 3.1000001.3
DEVICE_STATUS_BIT_FRC	BOOL	TRUE: Forced device	%MX 3.1000001.2
DEVICE_STATUS_BIT_DNR	BOOL	TRUE: Device not ready	%MX 3.1000001.1
DEVICE_STATUS_BIT_SER	BOOL	TRUE: System reserved	%MX 3.1000001.0

Table 65: MVB Device Status Word

Name	Type	Address
MVB_INITIALIZED	TYPE_Freshness	%MX 3.1000000.0

Table 66: Freshness

Definition of TYPE_Freshness:

```

TYPE
    TYPE_Freshness : ARRAY[0..4095] OF DINT;
END_TYPE

```

4.2. TRDP PD Communication - I303_DG_TRDP_DRV

The i303_DG_TRDP_DRV is the MULTIPROG Express' TRDP interface driver for TRDP (Train Real-Time Data Protocol) process data communication which utilises the i303 Ethernet module. Various steps must be processed in order to establish process data communication using TRDP. These steps are:

1. Define the Read and Write I/O groups for the TRDP application.
2. Select the "duagon TRDP Interface" as the I/O driver interface.
3. Define the I/O read-write variables for using the I/O groups and the I/O driver interface of MULTIPROG Express (henceforth simply referred to as MULTIPROG in section 4.2).
4. Download the TRDP configuration file i.e i303_trdp_config.xml along with the MULTIPROG TRDP application executable to the ionia™ with the help of the MULTIPROG Project Control Dialog.
5. Write the TRDP configuration file to the i303 Ethernet interface module (henceforth simply referred to as i303 in section 4.2).
6. Initialise the TRDP stack on the i303 with the i303_DG_TRDP_DRV I/O interface integrated in the ECLR application running on the ionia™.
7. Publish-Subscribe to the TRDP COM IDs specified in the TRDP application's I/O configuration.
8. Execute Read-Write operations with the help of the MULTIPROG Project Control Dialog.

4.2.1. TRDP Device Configuration

The TRDP device configuration is an XML configuration file which is parsed to initialize the TRDP stack on the i303 for specified Ethernet interfaces, TRDP COM IDs, IP addresses and data sets.

For the TRDP application, two devices communicating with each other need to have the same telegram and dataset definition for a given COM ID in order to exchange messages. The source and destination IP addresses for the telegrams being received or sent by the devices participating in the communication need to be customised depending on the Ethernet interfaces being used by the devices.

Application developers can refer "IEC61375-2-3:2015 Appendix C" for the configuration file description.

A sample TRDP XML configuration file named "i303_trdp_config.xml" is provided in the path "plc\eCLR\ARM_LE_GCC3" of the MULTIPROG installation directory. The telegram and dataset information provided in the configuration file can be used as a model to implement a customised TRDP xml configuration file. This file is also useful for utilising the example projects provided with the MULTIPROG installer.

4.2.2. Creating a MULTIPROG Express TRDP Application

There are some specifics that need to be attended to before creating a MULTIPROG TRDP application.

It should be verified that the "i303_DG_TRDP_DRV.DLL" dynamic link library for the i303_DG_TRDP_DRV interface driver is present in the "plc\eCLR\ARM_LE_GCC3" path of the MULTIPROG Express installation directory and a reference to that is present in the "eCLRDeviceProfiles.xml".

```
<Assembly Name="i101_DG_MVB_DRV" Version="0.0.1" IsNative="1" />
<Assembly Name="i301_DG_MVB_DRV" Version="0.0.1" IsNative="1" />
<Assembly Name="i103_DG_CO_DRV" Version="0.0.1" IsNative="1" />
<Assembly Name="i303_DG_TRDP_DRV" Version="0.0.1" IsNative="1" />
<Assembly Name="i306_DG_WTB_DRV" Version="0.0.1" IsNative="1" />
<Assembly Name="ionia_SM_DRV" Version="0.0.1" IsNative="1" />
<OnlineIncludeFile>i303_trdp.bin</OnlineIncludeFile>
<OnlineIncludeFile>i303_trdp_config.xml</OnlineIncludeFile>
```

Figure 22: Excerpts from eCLRDeviceProfiles.xml

In Figure 22, the "i303_DG_TRDP_DRV.DLL" is referenced. The "i303_trdp_config.xml" is also referenced. This is to ensure that the MULTIPROG TRDP application includes the i303_trdp_config.xml in the TRDP application executable that is to be executed by the ECLR application on ionia™.

For this the MULTIPROG project needs to be customised.

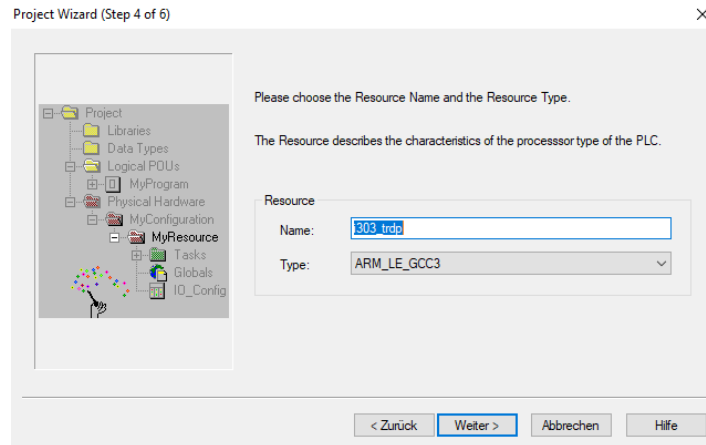


Figure 23: Naming the MULTIPROG TRDP application resource in project settings

As shown in Figure 23, the MULTIPROG TRDP application resource needs to be named "i303_trdp" in order for the i303_DG_TRDP_DRV driver in the ionia™ ECLR application to extract the TRDP XML configuration file for the i303 Ethernet module from the MULTIPROG TRDP application image.

As shown in Figure 24, in the "Resource Settings" of the MULTIPROG, application developers also need to specify the IP address of the ionia™ system with which the application will communicate, ECLR core build settings and the time-interval of read-write operations.

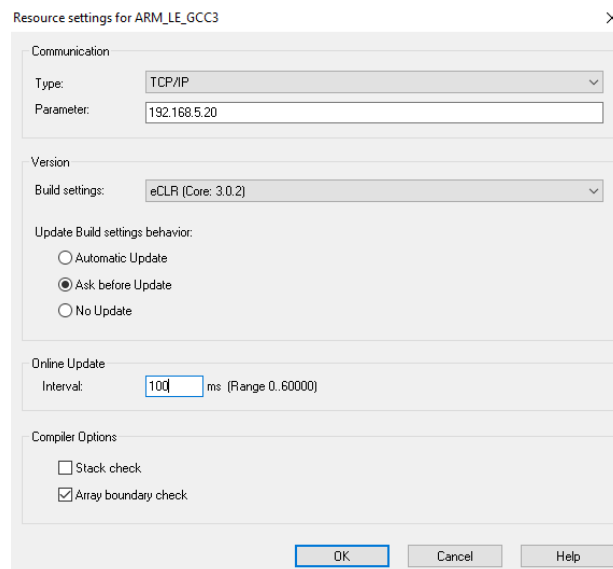


Figure 24: Example of Resource settings for MULTIPROG TRDP Application

The COM IDs to be used by the TRDP application need to be defined in MULTIPROG with the help of I/O group definitions required by MULTIPROG to execute read-write operations.

In MULTIPROG, the start addresses of an I/O group should serve as a TRDP COM ID. The I/O read-write addresses are 10-digit values which are bound by the unsigned 32-bit integer range as also the duagon TRDP API allows only unsigned 32-bit COM IDs.

As a result, application developers need to choose the application COM IDs from the unsigned 32-bit integer range (0 – 4294967295).

The TRDP application requires data packets of size 1432 bytes for its full operation. Hence, application developers need to be mindful of spacing the start addresses (and thereby the COM IDs) such that each COM ID has a data size of 1432 bytes allocated to it for read-write operations and that there is no overlap between the end address of a COM ID and the start address of the next COM ID. Thus, 4294965863 is the last COM ID value that can be used

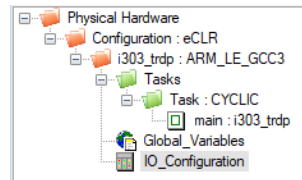


Figure 25: I/O Configuration functionality of MULTIPROG Express

Figure 25 highlights the I/O Configuration option which should be used to specify TRDP COM IDs are I/O groups in MULTIPROG.

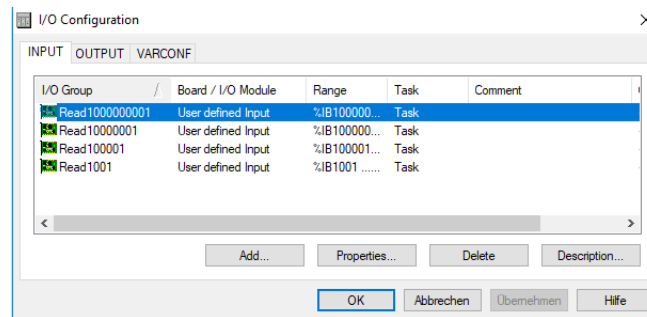


Figure 26: Example of INPUT groups of the MULTIPROG TRDP application

Figure 26 shows multiple INPUT groups specifying range of addresses to read from and the Task they are associated with it. Multiple INPUT groups can be created to read from specific addresses. They can be named as desired by the application developers. In this example, they are named as per the COM ID they are supposed to read from such as 1001, 100001, 100000001 and 1000000001. The Task is the top application which controls these I/O groups and is defined at the time of application creation. It can also be named as desired by the application developers. The OUTPUT group specifications can be similar as INPUT group specifications but need not be.

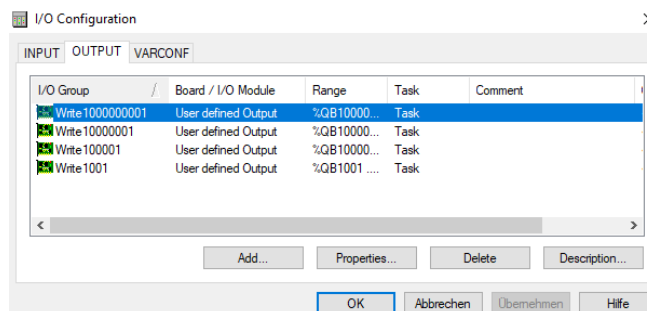


Figure 27: Example of OUTPUT groups of MULTIPROG Express application

The INPUT-OUTPUT groups specify the read-write locations of the MULTIPROG TRDP application. In these groups respectively, we specify starts and ends of read-write addresses. The start addresses correspond to the COM ID whereas the end addresses are separated from the start addresses by 1432 bytes making it possible to read-write a full TRDP data packet. If a write group is writing to a COM ID, application developers can create a read group reading from the same COM ID (start address offset).

Apart from specifying the COM IDs, the application developers also need to specify the driver to be used for the TRDP application. In the "Board I/O Module", application developers should select the "duagon i303 TRDP interface" and select "user defined Input", click on "Driver Parameter" and then specify "i303_DG_TRDP_DRV" in the "Driver Name" section, click on OK and then "Übernehmen" to apply changes to the I/O group. This procedure applies to both the INPUT and OUTPUT groups.

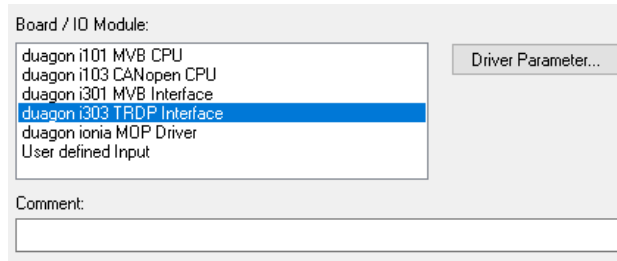


Figure 28: Example Input group configuration

Figure 28 shows an example configuration of a read group set up for COM ID 1000000001. The start address corresponds to the COM ID and the end address is the sum of the 1432-byte TRDP data packet size and the start address (1000001432). All read-write operations for COM ID 1000000001 must be done within this range.

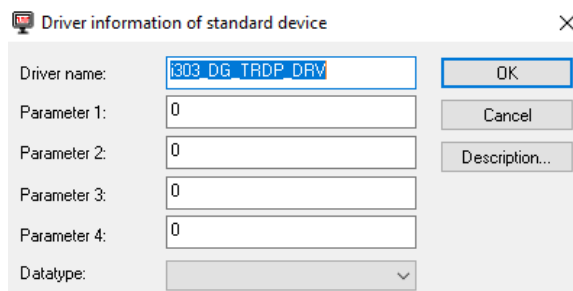


Figure 29: Specifying i303_DG_TRDP_DRV as the application's I/O interface

4.2.3. Defining I/O Variables for a Read-Write Application

Based on the I/O groups defined as per section 4.2.3, I/O variables should be defined as per the start and end addresses defined for a particular read-write group that has been configured to read messages from and write to a given COM ID. These I/O variables can be Boolean variables, signed and unsigned 8-bit, 16-bit, 32-bit integers, 32-bit Single and 64-bit Double integers.

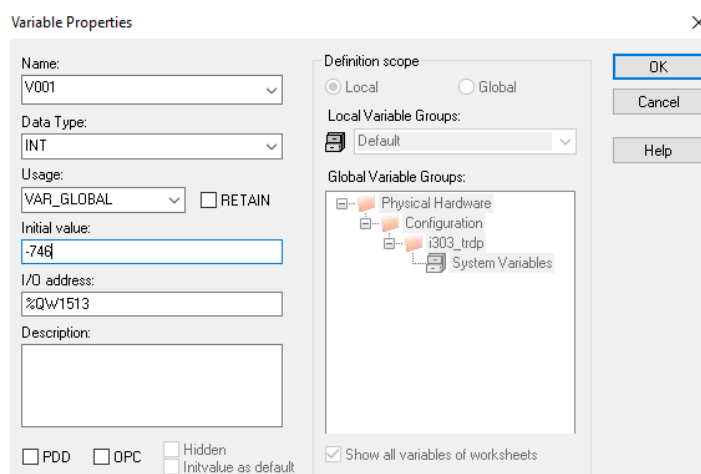


Figure 30: Example of variable declaration for COM ID 1001

Figure 30 shows declaration of a signed 16-bit integer V001 in MULTIPROG with an initial value of -746. The I/O address is specified as "%QW1513" where Q indicates an output resource, W indicates a 16-bit "WORD" and the decimal value 1513 is a sum of the COM ID 1001 and the offset 512 where the 16-bit integer is supposed to be written.

Figure 31 shows declaration of an unsigned 32-bit integer V029 which is used to store the value read from offset 0 of the COM ID. The I/O address is specified as "%ID100000001" where I indicates an input resource, D indicates 32-bit double "WORD" and the decimal value 100000001 is a sum of the COM ID and the offset 0 where the value is to be read from.

Such variables can be declared to create a complex MULTIPROG application.

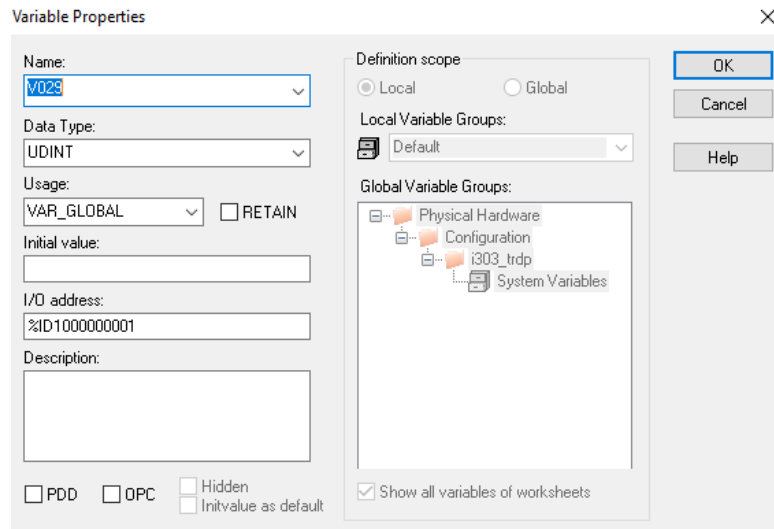


Figure 31: Example of variable declaration for COM ID 100000001

In order to declare and use the variables, the following IEC 61131 addressing format is presented in the context of the TRDP application:

% [I | Q] [X | B | W | D] [COM_ID + OFFSET]

Where:

- '%' is the header of an IEC address
- 'I' specifies an input resource
- 'Q' specifies an output resource
- 'X', 'B' 'W' and 'D' define, respectively, a BOOLEAN, an 8-bit integer, a 16-bits WORD and a 32-bits double WORD
- COM_ID + OFFSET defines the sum of the COM ID (start address) and the offset where the data is read from or written to for a given COM ID

4.2.4. Downloading the Application

Before downloading the application, the IP address of the target device should be verified in the Resource settings. The ECLR application with the i303 TRDP driver should be in a running state on the ionia™ at the time of downloading the ECLR application. On the ionia Linux CLI, users can run the ECLR application simply by typing "eCLR-App" and pressing "Enter".

```
root@ionia-softfp:~# eCLR-App
```

Figure 32: Executing ECLR application from ionia™ Linux CLI

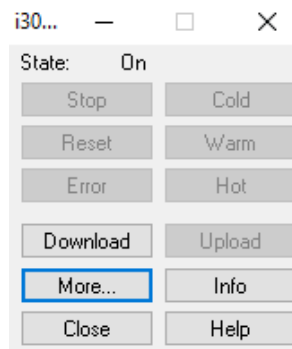


Figure 33: Project Control Dialog before application download

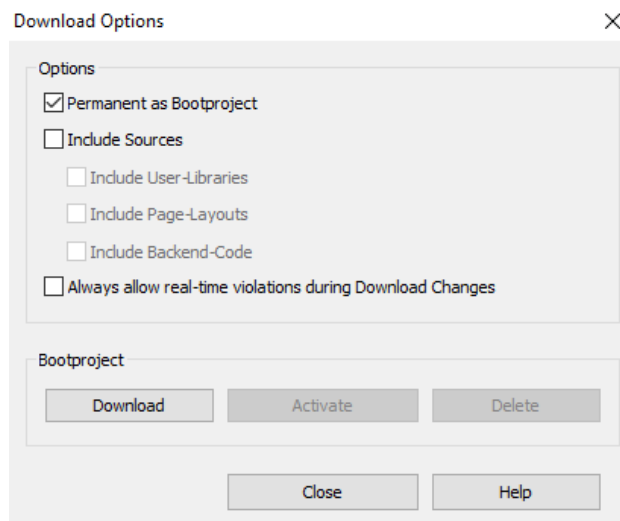


Figure 34: Download options for application download

Figures 33 and 34 show the Project Control Dialog before MULTIPROG TRDP application download. If any pre-existing MULTIPROG Express application is running on the ionia™ at the time of download, it should be removed by clicking "Reset" -> "Stop" -> "More..." -> "Delete". Then the MULTIPROG Express TRDP application can be downloaded by clicking "Download" under Bootproject section of "More...". The options under the Download Options can be selected as per the goals associated with the application.

Figure 35 shows how the MULTIPROG Project Control Dialog needs to be operated by the users when using the i303_DG_TRDP_DRV TRDP interface driver. It is important that the ECLR application is started on the ionia™, be it in whatever state, so that the MULTIPROG Project Control Dialog can help the user decide the actions that need to be taken. As long as the MULTIPROG Express application does not find the ECLR application in an accessible state, the Project Control Dialog displays a "Timeout" message with the buttons of the dialog box not accessible to the user, till the ECLR application becomes accessible once again. "Timeout" message is also displayed when a wrong IP address of the ionia™ is input in MULTIPROG application's Resource settings.

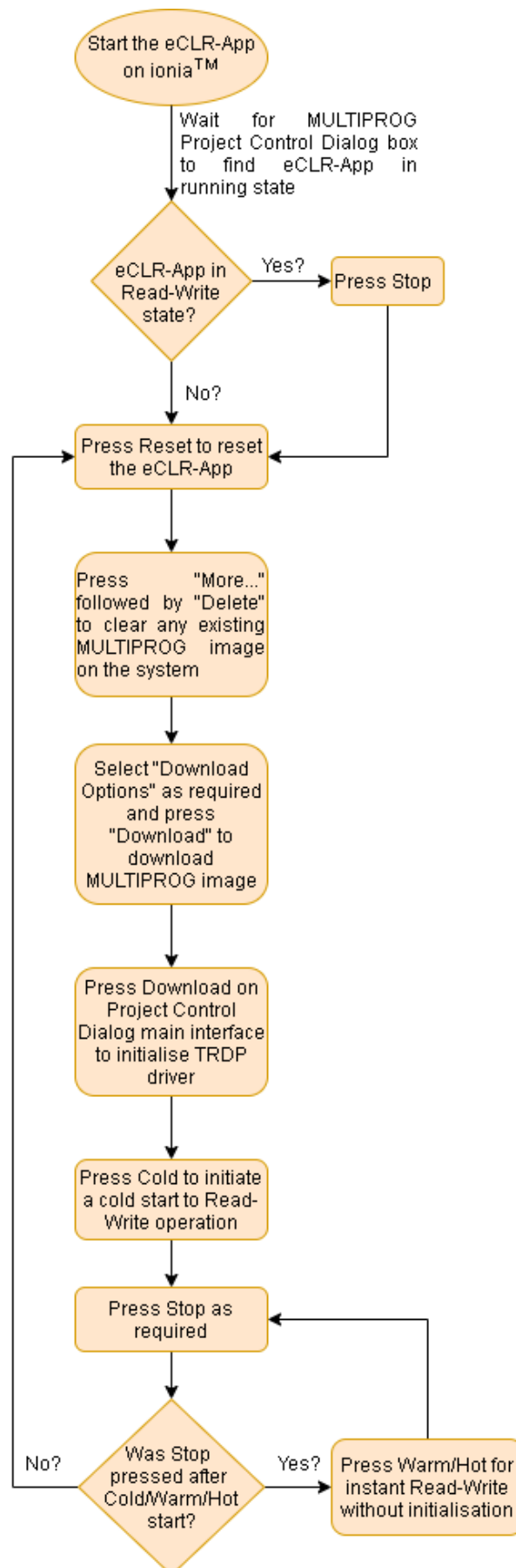


Figure 35: Instructions to use MULTIPROG Express Project Control Dialog

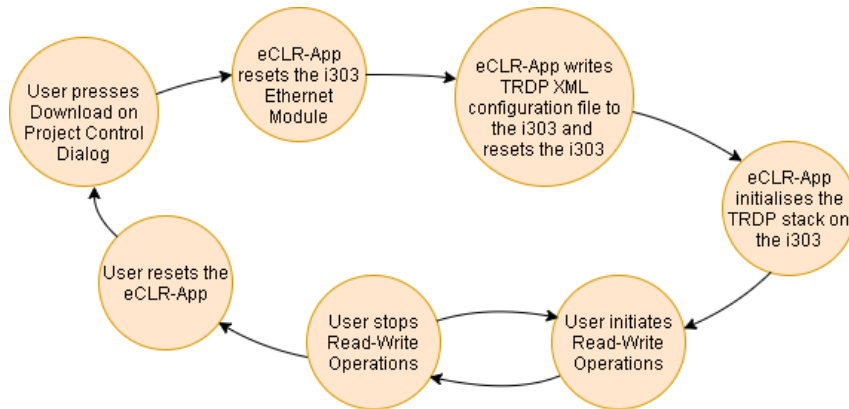


Figure 36: Operation cycle of i303_DG_TRDP_DRV as a part of the ECLR application in conjunction with the MULTIPROG TRDP application

Figure 36 shows the operation cycle of the i303_DG_TRDP_DRV TRDP interface driver as a part of the ECLR application. The following actions take place once the MULTIPROG TRDP application is downloaded onto ionia™:

- The ECLR application resets the i303 to ensure that the i303 is not in any old TRDP transactional state before a new TRDP transaction is initialised. The ECLR application continues further once it has identified that the i303's Ethernet interface needed for TRDP transactions is active.
- The ECLR application extracts the TRDP XML configuration file from the MULTIPROG TRDP application image and transfers it to the i303 and resets the i303 once again for letting the XML configuration take effect. The ECLR application continues further once it has identified that the i303's Ethernet interface needed for TRDP transactions is active.
- Then the ECLR application initialises the TRDP stack on the i303 for different COM IDs and datasets specified in the XML configuration before the TRDP read-write transactions can be initialised.
- The users can then initiate read-write transactions from the MULTIPROG Project Control Dialog using the "Cold" button on the dialog box.
- Remember that the "Cold" option can only be used in the beginning after the TRDP stack initialisation. Once the users stop the read-write operations by pressing "Stop" on the dialog box, in order to start Read-Write operations once again without stack initialisation, the "Warm" or "Hot" buttons on the dialog box should be used to initiate warm or hot read-write options (see Figure 35)
- Should the user desire to reset the MULTIPROG TRDP application, the user should first click "Stop" and then "Reset" button on the dialog box.
- To restart the TRDP application, the user should press "Download" on the dialog box and initiate the read-write operations as mentioned above.

4.2.5. Example Projects and Troubleshooting

Along with the MULTIPROG's revised installer (version d-010864-064840) 2 example projects, namely the i202_LED_Demo and i303_DG_TRDP_GetStarted are provided to help application developers get familiarized with MULTIPROG application development for i303_DG_TRDP_DRV driver. The i202_LED_Demo uses the values from i303_DG_TRDP_DRV to activate LED's on the i202 module. These projects should be unzipped using MULTIPROG and the example TRDP XML configuration file "i303_trdp_config.xml" should be copied from the path "plc\eCLR\ARM_LE_GCC3" of the MULTIPROG installation directory and placed in the path "C:\Configuration\R\i303_trdp" of both of these two unzipped projects' source directories. These source directories are found in the path "C:\Users\Public\Documents\MULTIPROG\Projects". Upon doing these, the projects should be built in the MULTIPROG and then used as demo applications.

If users experience any error with the MULTIPROG application the most basic way to troubleshoot this problem is to stop the MULTIPROG and the partner communication applications on the other end of the network. Then the ionia™ should be rebooted. Upon reboot, the MULTIPROG and the partner communication applications can be started and the TRDP communication can be established as before.

5. Constants and Error Codes

5.1. Address Constants

AM_SAME_STATION	= 0	/* This station, regardless of Station ID	*/
AM_UNKNOWN	= 255	/* Unknown Station ID	*/
AM_MAX_BUSSES	= 1	/* Maximum number of link layers supported	*/
		/* by an implementation [1..16]	*/
AM_ROUTER_FCT	= 254	/* Function ID of Router	*/
AM_AGENT_FCT	= 253	/* Function ID of Agent	*/
AM_MANAGER_FCT	= 254	/* Function ID of Manager	*/
AM_SAME_NODE	= 0	/* Communication under the same Node, not	*/
		/* going over the train bus	*/
AM_SYSTEM_ADDR	= 128	/* Bit 0 of the Node Address indicates a	*/
		/* System Address	*/
AM_ANY_TOPO	= 0	/* Top Counter in unknown	*/

5.2. Error Codes

5.2.1. Firmware Library Error Codes

DG_OK	0	/* No error	*/
DG_ERROR	400	/* General host driver error	*/
DG_ERROR_TIMEOUT	401	/* Timeout on read/write access	*/
DG_ERROR_COMM_FAIL	402	/* Communication failure	*/
DG_ERROR_OSL_SYNC_INIT	500	/* Mutex initialisation failed	*/
DG_ERROR_OSL_SYNC_LOCK	501	/* Mutex lock failed	*/
DG_ERROR_OSL_SYNC_UNLOCK	502	/* Mutex unlock failed	*/
DG_ERROR_OSL_SYNC_DESTROY	503	/* Mutex destroy failed	*/
DG_ERROR_OSL_INIT	520	/* I/O access initialisation failed	*/
DG_ERROR_OSL_READ	521	/* Read access failed	*/
DG_ERROR_OSL_WRITE	522	/* Write access failed	*/
DG_ERROR_OSL_NOT_READY	523	/* device registers not ready	*/
DG_ERROR_HDIO_RECV_WOULD_BLOCK	541	/* Receiving task would block	*/
DG_ERROR_HDIO_SEND_WOULD_BLOCK	542	/* Sending task would block	*/
DG_ERROR_HDIO_UNKNOWN_CHANNEL_TYPE	544	/* Unknown channel type	*/
DG_ERROR_HDIO_UNKNOWN_IO_MODE	545	/* I/O mode unknown	*/
DG_ERROR_HDIO_NOT_SUPPORTED	546	/* Not supported for config I/O mode	*/
DG_ERROR_HDC_PROTOCOL_MISMATCH	560	/* Received wrong protocol ID	*/
DG_ERROR_HDC_PROTOCOL	561	/* Error in HDC protocol	*/
DG_ERROR_HDC_HANDLE	562	/* Handle is wrong	*/
DG_ERROR_HDC_LENGTH_MISMATCH	563	/* Payload sz. Spec. in header n.a.	*/
DG_ERROR_RPC_UNKNOWN_TYPE	580	/* Unknown RPC type	*/
DG_ERROR_VERSION_DRIVER_OLD	590	/* host needs a newer device	*/
DG_ERROR_VERSION_DEVICE_OLD	591	/* device requires a newer host	*/
DG_ERROR_EMPTY_BLOCK	592	/* protocol not ready	*/
DG_ERROR_REGISTER_NOT_CORRECT	593	/* norm. means the dev has been reset	*/
DG_ERROR_FUNCTION_TIMED_OUT	594	/* time to execute func. exceeds limit	*/
DG_ERROR_CHANNEL_TIMED_OUT	595	/* previous function executed on this	*/
		/* channel exceeded timeout limit	*/
DG_ERROR_PARAMETER_EXCEEDS_BOUNDS	596	/* a range check has failed	*/
DG_ERROR_MODULE_CMD_NOT_EXIST	597	/* cmd does not exist on module	*/
DG_ERROR_CARD_NOT_FOUND	600	/* card not defined in dg_config.h	*/

EPERM	= 1	/* Operation not permitted	*/
ENOENT	= 2	/* No such file or directory	*/
ESRCH	= 3	/* No such process	*/
EINTR	= 4	/* Interrupted system call	*/
EIO	= 5	/* I/O error	*/
ENXIO	= 6	/* No such device or address	*/
E2BIG	= 7	/* Argument list too long	*/
ENOEXEC	= 8	/* Exec format error	*/
EBADF	= 9	/* Bad file number	*/
ECHILD	= 10	/* No child processes	*/
EAGAIN	= 11	/* Try again	*/
ENOMEM	= 12	/* Out of memory	*/
EACCES	= 13	/* Permission denied	*/
EFAULT	= 14	/* Bad address	*/
ENOTBLK	= 15	/* Block device required	*/
EBUSY	= 16	/* Device or resource busy	*/
EEXIST	= 17	/* File exists	*/
EXDEV	= 18	/* Cross-device link	*/
ENODEV	= 19	/* No such device	*/
ENOTDIR	= 20	/* Not a directory	*/
EISDIR	= 21	/* Is a directory	*/
EINVAL	= 22	/* Invalid argument	*/
ENFILE	= 23	/* File table overflow	*/
EMFILE	= 24	/* Too many open files	*/
ENOTTY	= 25	/* Not a typewriter	*/
ETXTBSY	= 26	/* Text file busy	*/
EFBIG	= 27	/* File too large	*/
ENOSPC	= 28	/* No space left on device	*/
ESPIPE	= 29	/* Illegal seek	*/
EROFS	= 30	/* Read-only file system	*/
EMLINK	= 31	/* Too many links	*/
EPIPE	= 32	/* Broken pipe	*/
EDOM	= 33	/* Math arg. out of domain of function	*/
ERANGE	= 34	/* Math result not representable	*/
EDEADLK	= 35	/* Resource deadlock would occur	*/
ENAMETOOLONG	= 36	/* File name too long	*/
ENOLCK	= 37	/* No record locks available	*/
ENOSYS	= 38	/* Function not implemented	*/
ENOTEMPTY	= 39	/* Directory not empty	*/
ELOOP	= 40	/* Too many symbolic links encountered	*/
EWouldBlock	= EAGAIN	/* Operation would block	*/
ENOMSG	= 42	/* No message of desired type	*/
EIDRM	= 43	/* Identifier removed	*/
ECHRNG	= 44	/* Channel number out of range	*/
EL2NSYNC	= 45	/* Level 2 not synchronized	*/
EL3HLT	= 46	/* Level 3 halted	*/
EL3RST	= 47	/* Level 3 reset	*/
ELNRNG	= 48	/* Link number out of range	*/
EUNATCH	= 49	/* Protocol driver not attached	*/
ENOSCSI	= 50	/* No CSI structure available	*/
EL2HLT	= 51	/* Level 2 halted	*/

EBADE	= 52	/* Invalid exchange	*/
EBADR	= 53	/* Invalid request descriptor	*/
EXFULL	= 54	/* Exchange full	*/
ENOANO	= 55	/* No anode	*/
EBADRQC	= 56	/* Invalid request code	*/
EBADSLT	= 57	/* Invalid slot	*/
EDEADLOCK	= EDEADLK		
EBFONT	= 59	/* Bad font file format	*/
ENOSTR	= 60	/* Device not a stream	*/
ENODATA	= 61	/* No data available	*/
ETIME	= 62	/* Timer expired	*/
ENOSR	= 63	/* Out of streams resources	*/
ENONET	= 64	/* Machine is not on the network	*/
ENOPKG	= 65	/* Package not installed	*/
EREMOTE	= 66	/* Object is remote	*/
ENOLINK	= 67	/* Link has been severed	*/
EADV	= 68	/* Advertise error	*/
ESRMNT	= 69	/* Srmount error	*/
ECOMM	= 70	/* Communication error on send	*/
EPROTO	= 71	/* Protocol error	*/
EMULTIHOP	= 72	/* Multihop attempted	*/
EDOTDOT	= 73	/* RFS specific error	*/
EBADMSG	= 74	/* Not a data message	*/
EOVERFLOW	= 75	/* Value too large for data type	*/
ENOTUNIQ	= 76	/* Name not unique on network	*/
EBADF	= 77	/* File descriptor in bad state	*/
EREMCHG	= 78	/* Remote address changed	*/
ELIBACC	= 79	/* Can't access a needed shared lib	*/
ELIBBAD	= 80	/* Accessing a corrupted shared lib	*/
ELIBSCN	= 81	/* .lib section in a.out corrupted	*/
ELIBMAX	= 82	/* Attempting to link in too many shared libs	*/
ELIBEXEC	= 83	/* Cannot exec a shared lib directly	*/
EILSEQ	= 84	/* Illegal byte sequence	*/
ERESTART	= 85	/* Interrupted system call should be restarted	*/
ESTRPIPE	= 86	/* Streams pipe error	*/
EUSERS	= 87	/* Too many users	*/
ENOTSOCK	= 88	/* Socket operation on non-socket	*/
EDESTADDRREQ	= 89	/* Destination address required	*/
EMSGSIZE	= 90	/* Message too long	*/
EPROTOTYPE	= 91	/* Protocol wrong type for socket	*/
ENOPROTOPT	= 92	/* Protocol not available	*/
EPROTONOSUPPORT	= 93	/* Protocol not supported	*/
ESOCKTNOSUPPORT	= 94	/* Socket type not supported	*/
EOPNOTSUPP	= 95	/* Operation not supported on transport endpoint	*/
EPFNOSUPPORT	= 96	/* Protocol family not supported	*/
EAFNOSUPPORT	= 97	/* Address family not supported by protocol	*/
EADDRINUSE	= 98	/* Address already in use	*/
EADDRNOTAVAIL	= 99	/* Cannot assign requested address	*/
ENETDOWN	= 100	/* Network is down	*/
ENETUNREACH	= 101	/* Network is unreachable	*/
ENETRESET	= 102	/* Network dropped connection because of reset	*/
ECONNABORTED	= 103	/* Software caused connection abort	*/
ECONNRESET	= 104	/* Connection reset by peer	*/
ENOBUFS	= 105	/* No buffer space available	*/
EISCONN	= 106	/* Transport endpoint is already connected	*/
ENOTCONN	= 107	/* Transport endpoint is not connected	*/
ESHUTDOWN	= 108	/* Cannot send after transport endpoint shutdown	*/
ETOOMANYREFS	= 109	/* Too many references: cannot splice	*/
ETIMEDOUT	= 110	/* Connection timed out	*/
ECONNREFUSED	= 111	/* Connection refused	*/
EHOSTDOWN	= 112	/* Host is down	*/
EHOSTUNREACH	= 113	/* No route to host	*/
EALREADY	= 114	/* Operation already in progress	*/
EINPROGRESS	= 115	/* Operation now in progress	*/
ESTALE	= 116	/* Stale NFS file handle	*/
EUCLEAN	= 117	/* Structure needs cleaning	*/
ENOTNAM	= 118	/* Not a XENIX named type file	*/
ENAVAIL	= 119	/* No XENIX semaphores available	*/
EISNAM	= 120	/* Is a named type file	*/
EREMOTEIO	= 121	/* Remote I/O error	*/
EDQUOT	= 122	/* Quota exceeded	*/
ENOMEDIUM	= 123	/* No medium found	*/
EMEDIUMTYPE	= 124	/* Wrong medium type	*/
ECANCELED	= 125	/* Operation Cancelled	*/
ENOKEY	= 126	/* Required key not available	*/
EKEYEXPIRED	= 127	/* Key has expired	*/
EKEYREVOKED	= 128	/* Key has been revoked	*/
EKEYREJECTED	= 129	/* Key was rejected by service	*/
EOWNERDEAD	= 130	/* Owner died	*/
ENOTRECOVERABLE	= 131	/* State not recoverable	*/
ERFKILL	= 132	/* Operation not possible due to RF-kill	*/
EHWPoison	= 133	/* Memory page has hardware error	*/

5.2.2. MVB Error Codes (AM_RESULT)

AM_OK	= 0	/* successful termination	*/
AM_FAILURE	= 1	/* unspecified failure	*/
AM_BUS_ERR	= 2	/* no bus transmission possible	*/
AM_REM_CONN_OVF	= 3	/* too many incoming connections	*/
AM_CONN_TMO_ERR	= 4	/* Connect_Request not answered	*/
AM_SEND_TMO_ERR	= 5	/* send time-out (connect was OK)	*/
AM_REPLY_TMO_ERR	= 6	/* no reply received	*/
AM_ALIVE_TMO_ERR	= 7	/* no complete message received	*/
AM_NO_LOC_MEM_ERR	= 8	/* not enough memory or timers	*/
AM_NO_REM_MEM_ERR	= 9	/* no more memory or timer (partner)	*/
AM_REM_CANC_ERR	= 10	/* cancelled by partner	*/
AM_ALREADY_USED	= 11	/* same operation already done	*/
AM_ADDR_FMT_ERR	= 12	/* address format error	*/
AM_NO_REPLY_EXP_ERR	= 13	/* no such reply expected	*/
AM_NR_OF_CALLS_OVF	= 14	/* too many calls requested	*/
AM_REPLY_LEN_OVF	= 15	/* Reply_Message too long	*/
AM_DUPL_LINK_ERR	= 16	/* duplicated conversation error	*/
AM_MY_DEV_UNKNOWN_ERR	= 17	/* my device unknown or not valid	*/
AM_NO_READY_INST_ERR	= 18	/* no ready Replier instance	*/
AM_NR_OF_INST_OVF	= 19	/* too many Replier instances	*/
AM_CALL_LEN_OVF	= 20	/* Call_Message too long	*/
AM_UNKNOWN_DEST_ERR	= 21	/* partner device unknown	*/
AM_INAUG_ERR	= 22	/* train inauguration occurred	*/
AM_TRY_LATER_ERR	= 23	/* (internally used only)	*/
AM_FIN_NOT_REG_ERR	= 24	/* final not registered	*/
AM_GW_FIN_NOT_REG_ERR	= 25	/* final not registered in router	*/
AM_GW_ORI_NOT_REG_ERR	= 26	/* origin not registered in router	*/
AM_MAX_ERR	= 31	/* highest system code	*/
		/* user codes are higher than	*/

5.2.3. i101 Error Codes

I101_OK	= 0	/* correct termination	*/
I101_DEVICENAME	= 1	/* Device Name plausibility Error	*/
I101_DEVICEADDRESS	= 2	/* Device Address plausibility Error	*/
I101_DEVICEBAENABLE	= 3	/* Bus administrator plausibility Error	*/
I101_RESOURCENAME	= 4	/* Resource Name plausibility Error	*/
I101_APPDOMAIN	= 10	/* Could not open application domain	*/
I101_APPFILECREATE	= 11	/* Could not create fd in app domain	*/
I101_APPFILEREAD	= 12	/* COULD NOT READ FILE FROM APP DOMAIN	*/
I101_FILEOPEN	= 13	/* COULD NOT CREATE FILE HANDLE	*/
I101_FILEWRITE	= 14	/* COULD NOT WRITE FILE	*/
I101_AP_RESULT	= 20		
I101_AS_RESULT	= 21		
I101_CM_RESULT	= 22		
I101_MM_RESULT	= 23		
I101_PVREAD	= 30	/* COULD NOT READ PV_NAME	*/
I101_PVWRITE	= 31	/* COULD NOT WRITE PV_NAME	*/
I101_UNSPECIFIED	= 90	/* Unspecified error	*/

5.2.4. IP Error Codes

#define IP_ENOERR	0	/* No error */	
#define IP_EPERM	1	/* Not permitted */	
#define IP_ENOENT	2	/* No such entity */	
#define IP_ESRCH	3	/* No such process */	
#define IP_EINTR	4	/* Operation interrupted */	
#define IP_EIO	5	/* I/O error */	
#define IP_EBADF	9	/* Bad file handle */	
#define IP_EAGAIN	11	/* Try again later */	
#define IP_EWOULDBLOCK	EAGAIN		
#define IP_ENOMEM	12	/* Out of memory */	
#define IP_EBUSY	16	/* Resource busy */	
#define IP_EXDEV	18	/* Cross-device link */	
#define IP_ENODEV	19	/* No such device */	
#define IP_ENOTDIR	20	/* Not a directory */	
#define IP_EISDIR	21	/* Is a directory */	
#define IP_EINVAL	22	/* Invalid argument */	
#define IP_ENFILE	23	/* Too many open files in system */	
#define IP_EMFILE	24	/* Too many open files */	
#define IP_EFBIG	27	/* File too large */	
#define IP_ENOSPC	28	/* No space left on device */	
#define IP_ESPIPE	29	/* Illegal seek */	
#define IP_EROFS	30	/* Read-only file system */	
#define IP_EDOM	33	/* Argument to math function outside valid domain*/	
#define IP_ERANGE	34	/* Math result cannot be represented */	
#define IP_EDEADLK	35	/* Resource deadlock would occur */	
#define IP_EDEADLOCK	IP_EDEADLK		
#define IP_ENOSYS	38	/* Function not implemented */	
#define IP_ENAMETOOLONG	60	/* File name too long */	
#define IP_ENOTEMPTY	66	/* Directory not empty */	
#define IP_ENOTSUP	95	/* Not supported error */	
#define IP_EEOF	200	/* End of file reached */	
#define IP_ENOSUPP	201	/* Operation not supported */	
#define IP_EDEVNOSUPP	202	/* Device does not support this operation */	

```

/* Additional errors used by networking */
#define IP_ENXIO      300 /* Device not configured */
#define IP_EACCES     301 /* Permission denied */
#define IP_EEXIST     302 /* File exists */
#define IP_ENOTTY     303 /* Inappropriate ioctl for device */
#define IP_EPIPE      304 /* Broken pipe */

/* non-blocking and interrupt i/o */
#define IP_EINPROGRESS 310 /* Operation now in progress */
#define IP_EALREADY    311 /* Operation already in progress */

/* ipc/network software - argument errors */
#define IP_ENOTSOCK    320 /* Socket operation on non-socket */
#define IP_EDESTADDRREQ 321 /* Destination address required */
#define IP_EMSGSIZE    322 /* Message too long */
#define IP_EPROTOTYPE  323 /* Protocol wrong type for socket */
#define IP_ENOPROTOOPT 324 /* Protocol not available */
#define IP_EPROTONOSUPPORT 325 /* Protocol not supported */
#define IP_ESOCKTNOSUPPORT 326 /* Socket type not supported */
#define IP_EOPNOTSUPP   327 /* Operation not supported */
#define IP_EPFNOSUPPORT 328 /* Protocol family not supported */
#define IP_EAFNOSUPPORT 329 /* Address family not supported by protocol */
#define IP_EADDRINUSE   330 /* Address already in use */
#define IP_EADDRNOTAVAIL 331 /* Can't assign requested address */
#define IP_EDATASIZE    332 /* Data size > 64k */

/* ipc/network software - operational errors */
#define IP_ENETDOWN     350 /* Network is down */
#define IP_ENETUNREACH  351 /* Network is unreachable */
#define IP_ENETRESET    352 /* Network dropped connection on reset */
#define IP_ECONNABORTED 353 /* Software caused connection abort */
#define IP_ECONNRESET   354 /* Connection reset by peer */
#define IP_ENOBUFS      355 /* No buffer space available */
#define IP_EISCONN      356 /* Socket is already connected */
#define IP_ENOTCONN     357 /* Socket is not connected */
#define IP_ESHUTDOWN    358 /* Can't send after socket shutdown */
#define IP_ETOOMANYREFS 359 /* Too many references: can't splice */
#define IP_ETIMEOUT     360 /* Operation timed out */
#define IP_ECONNREFUSED  361 /* Connection refused */
#define IP_EHOSTDOWN    364 /* Host is down */
#define IP_EHOSTUNREACH 365 /* No route to host */

```

5.2.5. i303_DG_TRDP_DRV Error Codes

```

#define DG_TRDP_NO_ERR      0 /* No error */
#define DG_TRDP_PARAM_ERR  -1 /* Parameter missing or out of range */
#define DG_TRDP_INIT_ERR   -2 /* Call without valid initialization */
#define DG_TRDP_NOINIT_ERR -3 /* Call with invalid handle */
#define DG_TRDP_TIMEOUT_ERR -4 /* Timeout */
#define DG_TRDP_NODATA_ERR -5 /* Non blocking mode: no data received */
#define DG_TRDP_SOCK_ERR   -6 /* Socket error / option not supported */
#define DG_TRDP_IO_ERR     -7 /* Socket IO error */
#define DG_TRDP_MEM_ERR    -8 /* No more memory available */
#define DG_TRDP_SEMA_ERR   -9 /* Semaphore not available */
#define DG_TRDP_QUEUE_ERR -10 /* Queue empty */
#define DG_TRDP_QUEUE_FULL_ERR -11 /* Queue full */
#define DG_TRDP_MUTEX_ERR -12 /* Mutex not available */
#define DG_TRDP_THREAD_ERR -13 /* Thread error */
#define DG_TRDP_BLOCK_ERR -14 /* System call would have blocked in blocking mode */
#define DG_TRDP_INTEGRATION_ERR -15 /* Alignment or endianness for selected target wrong */
#define DG_TRDP_NOCONN_ERR -16 /* No TCP connection */
#define DG_TRDP_NOSESSION_ERR -30 /* No such session */
#define DG_TRDP_SESSION_ABORT_ERR -31 /* Session aborted */
#define DG_TRDP_NOSUB_ERR -32 /* No subscriber */
#define DG_TRDP_NOPUB_ERR -33 /* No publisher */
#define DG_TRDP_NOLIST_ERR -34 /* No listener */
#define DG_TRDP_CRC_ERR -35 /* Wrong CRC */
#define DG_TRDP_WIRE_ERR -36 /* Wire */
#define DG_TRDP_TOPO_ERR -37 /* Invalid topo count */
#define DG_TRDP_COMID_ERR -38 /* Unknown ComId */
#define DG_TRDP_STATE_ERR -39 /* Call in wrong state */
#define DG_TRDP_APP_TIMEOUT_ERR -40 /* Application Timeout */
#define DG_TRDP_APP_REPLYTO_ERR -41 /* Application Reply Sent Timeout */
#define DG_TRDP_APP_CONFIRMTO_ERR -42 /* Application Confirm Sent Timeout */
#define DG_TRDP_REPLYTO_ERR -43 /* Protocol Reply Timeout */
#define DG_TRDP_CONFIRMTO_ERR -44 /* Protocol Confirm Timeout */
#define DG_TRDP_REQCONFIRMTO_ERR -45 /* Protocol Confirm Timeout (Request sender) */
#define DG_TRDP_PACKET_ERR -46 /* Incomplete message data packet */
#define DG_TRDP_UNRESOLVED_ERR -47 /* Unable to resolve hostname */
#define DG_TRDP_XML_PARSER_ERR -48 /* XML parser error */
#define DG_TRDP_INUSE_ERR -49 /* Resource is still in use */
#define DG_TRDP_MARSHALLING_ERR -50 /* Source size exceeded, dataset mismatch */
#define DG_TRDP_DG_MARSH_ERR -51 /* Marshalling: General error */
#define DG_TRDP_DG_MARSH_PARAM -52 /* Marshalling: Parameter error, e.g. null pointer */
#define DG_TRDP_DG_MARSH_MEM -53 /* Marshalling: Memory error */

```

```

#define DG_TRDP_DG_MARSH_INIT -54 /* Marshalling: Initialization error/not
initialized */
#define DG_TRDP_DG_MARSH_DSNOTFOUND -55 /* Marshalling: Dataset not found */
#define DG_TRDP_DG_MARSH_RECURSION -56 /* Marshalling: Recursion error -> max
recursion level reached */
#define DG_TRDP_DG_MARSH_BUFFSIZE -57 /* Marshalling: Provided buffer size is too
small */
#define DG_TRDP_DG_MARSH_CONFIG -58 /* Marshalling: Mal-configuration / or no
configuration at all. */
#define DG_TRDP_DG_MARSH_INVALID -59 /* Marshalling: Invalid function call at
this point */
#define DG_TRDP_STACK_NOT_READY -80 /* Stack is not yet fully initialized */
#define DG_TRDP_INVALID_FUNC -81 /* Invalid function call at this point */
#define DG_TRDP_NO_DNS_SERVER -82 /* Name resolution requested but no DNS
server available */
#define DG_TRDP_NO_FRESH_DATA -83 /* Valid subscriber found, but no updated
data since last read */
#define DG_TRDP_UNKNOWN_ERR -99 /* Unspecified error */

```

6. Links to other duagon Documents

In general, most of the documents are located on www.duagon.com, and may be downloaded from there in the most up to date version.

6.1. Software Documentation

D2000 Configuration Tool - Data Sheet, d-000341-nnnnnn

The document describes the usage of the duagon D2000 configuration tool. The tool allows to engineer all signals of a control system consisting of duagon target hardware (e.g. controllers, MVB interfaces, etc.) and compatible third-party target hardware.

6.2. User's Guides

Getting Started with ionia™ – User's Guide, d-007724-nnnnnn

This document describing first steps for set-up and configuration of the ionia™ system.

6.3. Tools

MULTIPROG Express Installer, d-007271-020555

Download link:

http://support.duagon.com/ionia/MULTIPROG_Express_d-007271-044329.exe

Necessary to install the MULTIPROG Express PLC programming tool together with all duagon libraries

MULTIPROG Express Installer, d-010864-064840

Download link:

http://support.duagon.com/ionia/MULTIPROG_Express_RED_d-010864-064840.exe

Necessary to install the MULTIPROG Express PLC programming tool together with all duagon libraries

D2000 Configuration Tool, d-000336-nnnnnn

duagon D2000 configuration tool.

7. Document History

d-006935-064706

- Added description on how to use the i303_DG_TRDP_DRV TRDP interface driver for the MULTIPROG Express controlled ECLR application on the ionia™.
- Added explanation of the TRDP driver operation cycle.
- Added information about example TRDP applications and basic troubleshooting
- Added TRDP Error Code description.
- Added installation instruction for the MULTIPROG Express version number d-010864-064840.

d-006935-022856

- Template updated.

d-006935-021080

- Socket errors added.

d-006935-020951

- Text corrections in i303 section.

d-006935-020675

- i211 module description updated.
- i303 module description added.
- DG_ERROR section updated.

d-006935-020293

- i201 and i202 module description updated.
- DG_ERROR section updated.

d-006935-018956

- ionia Redundancy Library added.
- i201 and i202 module description updated.
- DG_ERROR section updated.

d-006935-017420

- Chapter 1.1 Release Information: Updated.
- Chapter 2. FW Library: All tables updated regarding compatibility.
- Chapter 4 Duagon IO Driver: New.

d-006935-015200

- Preliminary release.
- Pictogram for i202_IN8 and i202_IN8_RB updated.

d-006935-014633

- Function blocks for library i102_DG_UDP_LIB now work with arrays as input/output instead of one data item accessed by an index.
- Library ionia_DG_FW_LIB for common ionia functionality added. Available: Function block for reading and setting the system time and date.
- MULTIPROG installer versions updated.

d-006935-014140

- The description of the FW library i20x_DG_FW_LIB is splitted off into independent chapters for each FW library; i.e. i201DG_FW_LIB , i202 DG_FW_LIB.
- Function blocks for library i201_DG_FW_LIB added.

- Function blocks for library i205_DG_FW_LIB added.
- Function blocks for library i101_DG_MVB_LIB added; i.e. MVB Message Data.
- Function blocks for library i102_DG_UDP_LIB added.
- Function blocks for library ionia_DG_FW_LIB added.
- CANopen Driver i103_DG_CO_DRV and its Driver Dialog i103_DG_CO_DLG added to chapter Revision Information.
- DG_ERROR code added.

d-006935-013206

- Initial version for G2.1 release.

Appendix A: Document Numbering System

All duagon documents have a unique identification number. The identification number has a certain internal structure to ease the tracking of different documents. In general, there are two parts:

Prefix	Document number	Filing number
d	-000310	-001952
Always constant	Specifies a certain purpose of a document with the intention to link several documents with different filing number. Please note, that the purpose of the document number is not stored for each document number, but can be derived from the document title, which is stored for each Filing number. The format is either 6 digits or not available.	Unique number, that identifies a document. Released in sequential manner as the documents are filed in the archive. A duagon-internal data base contains exactly one document title text for each filing number. Always 6 digits.

Table 67: Explanation of the Document Numbering System at duagon AG

Examples for identification numbers

Identification number	Document Title / Remarks
d-000310-001606 d-000310-001952	„DXIO data sheet Rev 2.2“ „DXIO data sheet Rev 2.3“ A document, that is updated from time to time: the document number has the purpose to link several versions of the „DXIO data sheet“ together. The filing number distinguishes between different versions. Please note, that the document number part is kept the same, if the basic intention of the early versions is kept, for example during revisions due to debugging or manufacturing updates. In case a significant change happens, another document number would be applied.
d-000719	„Notes from prototype meeting ...“ A document, that is obviously not updated after release. The „document number“ part is missing, and the filing number remains the only used part for identification.

Table 68: Examples for the Document Numbering System at duagon AG

Convention:

When a document is referenced by its document number, as for example "d-000584-nnnnnn", then the "newest" version of this specific document is meant. When a specific version of a document is intended, then the full identification number i.e. "d-000584-002043" is specified.

Appendix B: Software Licensing

The software components used for the FU and FBs are subject to specific license agreements. The following sections describe the relevant issues in a generalized form for duagon products:

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